Field data collection and data compilation

Analysing Ecological Data with Detection Error

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Preamble

```r
suppressPackageStartupMessages({
    library(ggplot2)
    library(corrplot)
    library(mefa4)
    library(skimr)
    library(dplyr)
    library(ggplot2)
    theme_set(theme_light())
    library(GGally)
    library(partykit)
})
```

ABMI data

Raw data

We start by loading the raw data file and taking a first look at its contents. `read.csv()` imports the ABMI breeding birds table into `b`. `str(b)` then prints a compact structural overview — number of rows and columns, each variable's name and type, and a preview of the first few values — which is the fastest way to confirm that columns were parsed correctly on import. `skimr::skim(b)` follows with a richer per-column summary: missing value counts, completeness rates, and type-appropriate statistics such as min/max/mean for numerics and frequency counts for characters.

```r
if (!interactive()) {
    b <- read.csv("../data/A_T26A_Breeding_Birds_AllYear_AB.csv")
} else {
    b <- read.csv("data/A_T26A_Breeding_Birds_AllYear_AB.csv")
}

str(b)
```

```
'data.frame':   213708 obs. of  22 variables:
 $ Rotation                              : chr  "Prototype" "Prototype" "Prototype" "Prototyp
 $ ABMI.Site                             : chr  "630" "630" "630" "630" ...
 $ Year                                  : int  2003 2003 2003 2003 2003 2003 2003 2003 2003
 $ Field.Date                            : chr  "13-Jun-03" "13-Jun-03" "13-
Jun-03" "13-Jun-03" ...
 $ Field.Crew.Member.s.                  : chr  "CS" "CS" "CS" "CS" ...
 $ Point.Count.Station                   : int  1 1 1 1 1 1 1 1 2 2 ...
```

```
 $ Identification.Date                   : chr  "VNA" "VNA" "VNA" "VNA" ...
 $ Identification.Analyst                : chr  "TAH" "TAH" "TAH" "TAH" ...
 $ Wind.Conditions                       : chr  "3" "3" "3" "3" ...
 $ Precipitation                         : chr  "None" "None" "None" "None" ...
 $ Start.of.Point.Count..24.hour.clock.  : chr  "5:17" "5:17" "5:17" "5:17" ...
 $ End.of.Point.Count..24.hr.clock.      : chr  "DNC" "DNC" "DNC" "DNC" ...
 $ Common.Name                           : chr  "Red-eyed Vireo" "Rose-
breasted Grosbeak" "Ovenbird" "Ovenbird" ...
 $ Scientific.Name                       : chr  "Vireo olivaceus" "Pheucticus ludovicianus" "
 $ Taxonomic.Resolution                  : chr  "Species" "Species" "Species" "Species" ...
 $ Unique.Taxonomic.Identification.Number: chr  "179021" "179139" "726205" "726205" ...
 $ Time.First.Detected..seconds.         : chr  "260" "40" "20" "50" ...
 $ Interval.1..0.200.seconds.            : chr  "PNA" "PNA" "PNA" "PNA" ...
 $ Interval.2..201.400.seconds.          : chr  "PNA" "PNA" "PNA" "PNA" ...
 $ Interval.3..401.600.seconds.          : chr  "PNA" "PNA" "PNA" "PNA" ...
 $ Behaviour                             : chr  "Singing" "Singing" "Singing" "Singing" ...
 $ X                                     : logi  NA NA NA NA NA NA ...
```

```r
skimr::skim(b)
```

Table 1: Data summary

Name	b
Number of rows	213708
Number of columns	22
Column type frequency:	
character	19
logical	1
numeric	2
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Rotation	0	1	9	10	0	3	0
ABMI.Site	0	1	1	15	0	1246	0
Field.Date	0	1	3	9	0	298	0
Field.Crew.Member.s.	0	1	2	11	0	263	0

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Identification.Date	0	1	3	11	0	819	0
Identification.Analyst	0	1	2	8	0	27	0
Wind.Conditions	0	1	1	3	0	8	0
Precipitation	0	1	3	7	0	7	0
Start.of.Point.Count..24.hour.clock.	0	1	2	5	0	847	0
End.of.Point.Count..24.hr.clock.	0	1	3	5	0	44	0
Common.Name	0	1	3	48	0	270	0
Scientific.Name	0	1	3	29	0	270	0
Taxonomic.Resolution	0	1	3	9	0	5	0
Unique.Taxonomic.Identification.Number	0	1	3	8	0	269	0
Time.First.Detected..seconds.	0	1	1	4	0	587	0
Interval.1..0.200.seconds.	0	1	1	4	0	6	0
Interval.2..201.400.seconds.	0	1	1	4	0	6	0
Interval.3..401.600.seconds.	0	1	1	4	0	6	0
Behaviour	0	1	3	41	0	22	0

Variable type: logical

skim_variable	n_missing	complete_rate	mean	count
X	213708	0	NaN	:

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
Year	0	1	2010.74	3.08	2003	2009	2011	2013	2015	
Point.Count.Station	0	1	4.83	2.55	1	3	5	7	9	

A tidy workflow would involve the following steps:

- set the survey metadata table (site, year, coordinates, etc.)
- join the predictor variables (climate, land cover, human footprint) using dplyr's `left_join()` by site and year or using `match()` to align rows.
- join the response variables (bird counts), 1 species at a time

Alternatively, we can pivot the bird counts to a wide format, where each species is a column, and then join this data to the the predictors. Here is an example of how to do this with the `Xtab` function from the `mefa4` package, which creates a cross-tabulation of counts by site and species. The resulting object `y` will have sites as rows and species as columns, with the counts of each species at each site.

```
y <- mefa4::Xtab(~ ABMI.Site + Common.Name, b)

head(y)
```

```
6 x 270 sparse Matrix of class "dgCMatrix"

  [[ suppressing 270 column names 'Accipiters', 'Alder Flycatcher', 'American Avocet' ... ]]


1    . . .
```

```
1008 . . . . . .  . . . . . 2 . . . 7 . . . 4

1    . . . . . 2 . . 7 . . . . . .
1001 . . . . . 1 . . . . . 3 . . . . . 1 . 3 . . . .
1002 . . . . . 1 . . . . . 2 . 1 . .
1003 . . . . . 1 . . . . . 6 . . . .
1007 . . . . . 1 . 4 . . . . . .
1008 . . . . . 6 . 4 . . . 1 . 3 . . . .

1    . 2 . 2 . . . . . 10 . . . . . 1 . . . .
1001 . 3 . . . . . 2 . . . . . 1
1002 . 4 . . . . . 11 . . . . . .
1003 . 2 . . . . . 8 . . . . . .
1007 . 2 2 . . . . . 6 . . . . . 2 . . . .
1008 . 6 3 1 . . . . . 7 . . 1 . . . . . .

1    . . . . . 3 . . . . . 1 .
1001 . . . . . 9 . . . . . 9 . . . . . 5 . . . . 1 . . 4 .
1002 . . . . . 2 . . . . . 1 . . . 2
1003 . . . . . 1
1007 . . . . . 9 . . . . . 3 . . . 4 . . . 9 4
1008 . . . . . 27 . . 6 . . . . . 10 . . . 1 . . . 18 2

1    . . . . . 11 .
1001 . . 7 2 . . 2 . . . 2 . 8 .
1002 . . 1 . . . 1 . . . . . 15 .
1003 . . . 1 . . 2 . . . . . 18 .
1007 . . . . . 31 .
1008 . . 2 . 2 . 2 . 1 . 2 . 19 1
```

The object `y` is a so-called sparse matrix, which is a memory-efficient way to store data with many zeros (as is common in ecological count data). Use `as.matrix()` to convert it to a regular matrix if needed, e.g. before joining to the data frame with the other site variables.

Cleaned data

In the next chunk, we switch from the raw table to a cleaned, analysis-ready data set.

First, we read the `ab_birds_site_year` object from disk. This object stores the bird counts and the site-level variables in a format that is easier to work with than the original CSV. It also contains the data in a way that is more suitable for modeling, with separate matrices for counts and predictors. The `d` object is a list that contains at least two components: `counts`,

which is a matrix of bird counts for each species, and `vars`, which is a data frame of site-year predictors.

Next, we choose one focal species by setting `spp` to `"yellowrumped_warbler"`. This lets us pull out a single response variable from the full community data set.

Finally, we create a new data frame called `x`. The column `y` contains the counts for the selected species, and the remaining columns come from `d$vars`, which holds the explanatory variables for each site-year. This gives us one table with the response and predictors side by side, ready for plotting or modeling.

```r
if (!interactive()) {
    d <- readRDS("../data/ab_birds_site_year.rds")
} else {
    d <- readRDS("data/ab_birds_site_year.rds")
}

spp <- "yellowrumped_warbler"

x <- data.frame(y = d$counts[, spp], d$vars)

x$yday <- as.POSIXlt(x$fielddate)$yday / 365
x$hr <- x$fieldtime / 24
```

Variables:

- Metadata:
  - `site`, `year`, `site_year`: site, year, unique ID for site/year (879 sites, 7911 points, 2003–2014),
  - `latitude`, `longitude`: public coordinates (decimal degrees)
  - `nrname`, `nsrname`: natural regions/subregions
- Bioclimatic variables
  - `ahm`: annual heat-moisture index (scaled temp/precip),
  - `pet`: potential evapotranspiration,
  - `ffp`: frost free period
  - `map`: mean annual precipitaton,
  - `mat`: mean annual temperature,
  - `mcmt`: mean cold month (January) temperature,
  - `mwmt`: mean warm month (July) temperature.
- Land cover: natural vegetation (150 m radius circular buffer at points, averaged at sites, given as proportion [0–1] of total 7.1 ha buffer area)

  - `pforest`: proportion of forest (coniferous, mixedwood, deciduous, Treed fens and bogs, includes cutblocks),
  - `pconif`: proportion of coniferous forest (upland and lowland, includes cutblocks),
  - `pwet`: proportion of lowland (wet) areas (treed and non-treed),
  - `poldfor`: proportion of mature and old-growth (>80 yrs) forest.

- Land cover: human footprint (0-1 as well as natural vegetation)
  - `cult`: cultivation,
  - `urbind`: urban/industrial,
  - `softlin`: 'soft' (vegetated) linear features (seismic lines, pipelines, power lines, road verges),
  - `hardlin`: 'hard' (non-vegetated) linear features (road and rails),
  - `cc`: forestry cutblocks,
  - `alien`: 'alienating' disturbances (soil disturbed, non regenerating, `cult + urbind + hardlin`),
  - `succ`: 'successional' disturbances (soil undisturbed, regenerating, `cc + softlin`),
  - `thf`: total human footprint (`alien + succ`).

Data exploration

Overview of the data frame:

```r
head(x) # try also tail(x)
```

```
           y site_year site year longitude latitude          nrname
1_2009    11    1_2009    1 2009 -110.2824 59.94313 Canadian Shield
1001_2011  0 1001_2011 1001 2011 -118.1261 54.31477       Foothills
1002_2011 15 1002_2011 1002 2011 -117.8462 54.29271       Foothills
1003_2011 18 1003_2011 1003 2011 -117.5166 54.26610       Foothills
1007_2008 31 1007_2008 1007 2008 -116.2694 54.18933       Foothills
1008_2008 19 1008_2008 1008 2008 -115.9910 54.14914       Foothills
                  nsrname  ahm pet ffp map  mat  mcmt mwmt   pforest    pconif
1_2009      Kazan Uplands 17.8 367  97 354 -3.7 -25.6 15.2 0.9498135 0.4740256
1001_2011 Upper Foothills 17.2 476 109 711  2.2  -9.5 13.9 0.9869118 0.9869118
1002_2011 Upper Foothills 17.4 474 107 700  2.2  -9.8 13.9 0.9158207 0.9158207
1003_2011 Upper Foothills 17.9 482 108 683  2.2 -10.1 14.0 0.9759079 0.9759079
1007_2008 Lower Foothills 20.7 531 105 590  2.2 -12.7 15.1 0.9474650 0.7056995
1008_2008 Lower Foothills 21.0 537 105 577  2.1 -13.0 15.3 0.9234332 0.4200765
                pwet    poldfor cult      urbind     softlin hardlin
1_2009    0.01944699 0.00000000    0 0.000000000 0.000000000       0
1001_2011 0.02386701 0.49759870    0 0.000000000 0.004490133       0
```

```
1002_2011 0.63633462 0.91292200    0 0.002696789 0.053533681       0
1003_2011 0.52992410 0.97590795    0 0.000000000 0.024092054       0
1007_2008 0.25240993 0.01826507    0 0.020364729 0.022357052       0
1008_2008 0.34410014 0.33486650    0 0.000000000 0.053046864       0
                   cc       alien       succ        thf      wind precip
1_2009    0.000000000 0.000000000 0.00000000 0.00000000 3.3333333   None
1001_2011 0.489313137 0.000000000 0.49380327 0.49380327 1.2222222   None
1002_2011 0.002898737 0.002696789 0.05643242 0.05912921 1.5555556   None
1003_2011 0.000000000 0.000000000 0.02409205 0.02409205 0.0000000   None
1007_2008 0.000000000 0.020364729 0.02235705 0.04272178 1.1111111   None
1008_2008 0.246410775 0.000000000 0.29945764 0.29945764 0.6666667   None
           fielddate fieldtime      yday        hr
1_2009    2009-06-11  7.881481 0.4410959 0.3283951
1001_2011 2011-06-13  8.038889 0.4465753 0.3349537
1002_2011 2011-06-09  8.194444 0.4356164 0.3414352
1003_2011 2011-06-10  8.218519 0.4383562 0.3424383
1007_2008 2008-05-28  9.157407 0.4054795 0.3815586
1008_2008 2008-06-01  7.212963 0.4164384 0.3005401
```

```r
str(x)
```

```
'data.frame':   837 obs. of  33 variables:
 $ y        : num  11 0 15 18 31 19 6 0 0 0 ...
 $ site_year: Factor w/ 879 levels "630_2003","656_2003",..: 180 320 321 322 110 111 112 323
 $ site     : Factor w/ 879 levels "1","1001","1002",..: 1 2 3 4 5 6 7 8 9 10 ...
 $ year     : int  2009 2011 2011 2011 2008 2008 2008 2011 2011 2011 ...
 $ longitude: num  -110 -118 -118 -118 -116 ...
 $ latitude : num  59.9 54.3 54.3 54.3 54.2 ...
 $ nrname   : Factor w/ 6 levels "Boreal","Canadian Shield",..: 2 3 3 3 3 3 1 5 1 1 ...
 $ nsrname  : Factor w/ 21 levels "Alpine","Athabasca Plain",..: 10 21 21 21 12 12 4 5 7 7 .
 $ ahm      : num  17.8 17.2 17.4 17.9 20.7 ...
 $ pet      : int  367 476 474 482 531 537 554 556 558 566 ...
 $ ffp      : int  97 109 107 108 105 105 104 112 113 112 ...
 $ map      : int  354 711 700 683 590 577 563 484 460 445 ...
 $ mat      : num  -3.7 2.2 2.2 2.2 2.2 ...
 $ mcmt     : num  -25.6 -9.5 -9.8 -10.1 -12.7 ...
 $ mwmt     : num  15.2 13.9 13.9 14 15.1 ...
 $ pforest  : num  0.95 0.987 0.916 0.976 0.947 ...
 $ pconif   : num  0.474 0.987 0.916 0.976 0.706 ...
 $ pwet     : num  0.0194 0.0239 0.6363 0.5299 0.2524 ...
 $ poldfor  : num  0 0.4976 0.9129 0.9759 0.0183 ...
 $ cult     : num  0 0 0 0 0 ...
```

```
$ urbind   : num  0 0 0.0027 0 0.0204 ...
$ softlin  : num  0 0.00449 0.05353 0.02409 0.02236 ...
$ hardlin  : num  0 0 0 0 0 ...
$ cc       : num  0 0.4893 0.0029 0 0 ...
$ alien    : num  0 0 0.0027 0 0.0204 ...
$ succ     : num  0 0.4938 0.0564 0.0241 0.0224 ...
$ thf      : num  0 0.4938 0.0591 0.0241 0.0427 ...
$ wind     : num  3.33 1.22 1.56 0 1.11 ...
$ precip   : Factor w/ 5 levels "None","Drizzle",..: 1 1 1 1 1 1 1 1 1 1 ...
$ fielddate: Date, format: "2009-06-11" "2011-06-13" ...
$ fieldtime: num  7.88 8.04 8.19 8.22 9.16 ...
$ yday     : num  0.441 0.447 0.436 0.438 0.405 ...
$ hr       : num  0.328 0.335 0.341 0.342 0.382 ...
```

```r
summary(x)
```

```
      y             site_year        site          year        longitude
 Min.   : 0.000   630_2003:  1   1      :  1   Min.   :2003   Min.   :-120.0
 1st Qu.: 0.000   656_2003:  1   1001   :  1   1st Qu.:2009   1st Qu.:-115.7
 Median : 1.000   661_2003:  1   1002   :  1   Median :2011   Median :-113.1
 Mean   : 4.528   717_2003:  1   1003   :  1   Mean   :2011   Mean   :-113.8
 3rd Qu.: 8.000   718_2003:  1   1007   :  1   3rd Qu.:2013   3rd Qu.:-111.5
 Max.   :38.000   790_2003:  1   1008   :  1   Max.   :2014   Max.   :-110.0
                  (Other) :831   (Other):831
   latitude                 nrname                       nsrname
 Min.   :49.04   Boreal         :429   Central Mixedwood     :189
 1st Qu.:51.72   Canadian Shield: 20   Dry Mixedgrass        :105
 Median :54.43   Foothills      : 77   Dry Mixedwood         :105
 Mean   :54.36   Grassland      :218   Central Parkland      : 69
 3rd Qu.:56.84   Parkland       : 78   Lower Boreal Highlands: 59
 Max.   :60.01   Rocky Mountain : 15   Lower Foothills       : 44
                                        (Other)               :266
     ahm             pet             ffp             map
 Min.   : 6.10   Min.   :326.0   Min.   : 70.0   Min.   : 272.0
 1st Qu.:20.50   1st Qu.:501.0   1st Qu.: 97.0   1st Qu.: 390.0
 Median :23.10   Median :538.0   Median :105.0   Median : 439.0
 Mean   :26.15   Mean   :547.1   Mean   :104.4   Mean   : 452.1
 3rd Qu.:30.50   3rd Qu.:606.0   3rd Qu.:113.0   3rd Qu.: 482.0
 Max.   :50.20   Max.   :738.0   Max.   :130.0   Max.   :1360.0

     mat              mcmt            mwmt         pforest
 Min.   :-4.600   Min.   :-26.9   Min.   : 9.9   Min.   :0.0000
```

```
 1st Qu.:-0.400   1st Qu.:-21.0   1st Qu.:15.3   1st Qu.:0.0000
 Median : 1.600   Median :-15.5   Median :16.0   Median :0.4347
 Mean   : 1.266   Mean   :-16.3   Mean   :16.1   Mean   :0.4461
 3rd Qu.: 2.900   3rd Qu.:-12.0   3rd Qu.:16.9   3rd Qu.:0.8694
 Max.   : 5.500   Max.   : -7.9   Max.   :19.7   Max.   :1.0000

     pconif             pwet            poldfor             cult
 Min.   :0.00000   Min.   :0.00000   Min.   :0.00000   Min.   :0.0000
 1st Qu.:0.00000   1st Qu.:0.00000   1st Qu.:0.00000   1st Qu.:0.0000
 Median :0.03202   Median :0.05258   Median :0.01054   Median :0.0000
 Mean   :0.27165   Mean   :0.22664   Mean   :0.20867   Mean   :0.2762
 3rd Qu.:0.56788   3rd Qu.:0.38026   3rd Qu.:0.38235   3rd Qu.:0.6407
 Max.   :1.00000   Max.   :1.00000   Max.   :1.00000   Max.   :1.0000

     urbind           softlin            hardlin               cc
 Min.   :0.00000   Min.   :0.000000   Min.   :0.000000   Min.   :0.00000
 1st Qu.:0.00000   1st Qu.:0.001997   1st Qu.:0.000000   1st Qu.:0.00000
 Median :0.00000   Median :0.016211   Median :0.000000   Median :0.00000
 Mean   :0.01432   Mean   :0.023343   Mean   :0.004301   Mean   :0.03211
 3rd Qu.:0.01070   3rd Qu.:0.034514   3rd Qu.:0.002997   3rd Qu.:0.00000
 Max.   :0.85719   Max.   :0.195696   Max.   :0.111614   Max.   :0.95381

     alien             succ               thf               wind
 Min.   :0.0000   Min.   :0.000000   Min.   :0.00000   Min.   :0.0000
 1st Qu.:0.0000   1st Qu.:0.002869   1st Qu.:0.01476   1st Qu.:0.6667
 Median :0.0148   Median :0.020173   Median :0.12274   Median :1.2222
 Mean   :0.2948   Mean   :0.055458   Mean   :0.35029   Mean   :1.4264
 3rd Qu.:0.6994   3rd Qu.:0.044361   3rd Qu.:0.75359   3rd Qu.:2.0000
 Max.   :1.0000   Max.   :0.963890   Max.   :1.00000   Max.   :5.0000

     precip      fielddate            fieldtime           yday
 None   :786   Min.   :2003-05-28   Min.   : 4.424   Min.   :0.3534
 Drizzle: 31   1st Qu.:2009-06-10   1st Qu.: 6.826   1st Qu.:0.4082
 Rain   : 10   Median :2011-06-15   Median : 7.211   Median :0.4192
 Fog    :  9   Mean   :2011-04-20   Mean   : 7.317   Mean   :0.4277
 Snow   :  1   3rd Qu.:2013-06-01   3rd Qu.: 7.739   3rd Qu.:0.4493
               Max.   :2014-06-24   Max.   :14.539   Max.   :0.6384

       hr
 Min.   :0.1843
 1st Qu.:0.2844
 Median :0.3005
 Mean   :0.3049
```

```
 3rd Qu.:0.3225
 Max.   :0.6058
```

```r
skimr::skim(x)
```

Table 5: Data summary

Name	x
Number of rows	837
Number of columns	33
Column type frequency:	
Date	1
factor	5
numeric	27
Group variables	None

Variable type: Date

skim_variable	n_missing	complete_rate	min	max	median	n_unique
fielddate	0	1	2003-05-28	2014-06-24	2011-06-15	203

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
site_year	0	1	FALSE	837	630: 1, 656: 1, 661: 1, 717: 1
site	0	1	FALSE	837	1: 1, 100: 1, 100: 1, 100: 1
nrname	0	1	FALSE	6	Bor: 429, Gra: 218, Par: 78, Foo: 77
nsrname	0	1	FALSE	21	Cen: 189, Dry: 105, Dry: 105, Cen: 69
precip	0	1	FALSE	5	Non: 786, Dri: 31, Rai: 10, Fog: 9

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
y	0	1	4.53	6.16	0.00	0.00	1.00	8.00	38.00	
year	0	1	2010.87	2.57	2003.00	2009.00	2011.00	2013.00	2014.00	
longitude	0	1	-113.77	2.72	-120.04	-115.69	-113.12	-111.47	-110.00	
latitude	0	1	54.36	3.00	49.04	51.72	54.43	56.84	60.01	
ahm	0	1	26.15	8.43	6.10	20.50	23.10	30.50	50.20	
pet	0	1	547.14	79.28	326.00	501.00	538.00	606.00	738.00	
ffp	0	1	104.39	10.63	70.00	97.00	105.00	113.00	130.00	
map	0	1	452.11	104.61	272.00	390.00	439.00	482.00	1360.00	
mat	0	1	1.27	2.26	-4.60	-0.40	1.60	2.90	5.50	
mcmt	0	1	-16.30	4.95	-26.90	-21.00	-15.50	-12.00	-7.90	
mwmt	0	1	16.10	1.48	9.90	15.30	16.00	16.90	19.70	
pforest	0	1	0.45	0.41	0.00	0.00	0.43	0.87	1.00	
pconif	0	1	0.27	0.35	0.00	0.00	0.03	0.57	1.00	
pwet	0	1	0.23	0.31	0.00	0.00	0.05	0.38	1.00	
poldfor	0	1	0.21	0.30	0.00	0.00	0.01	0.38	1.00	
cult	0	1	0.28	0.38	0.00	0.00	0.00	0.64	1.00	
urbind	0	1	0.01	0.05	0.00	0.00	0.00	0.01	0.86	
softlin	0	1	0.02	0.03	0.00	0.00	0.02	0.03	0.20	
hardlin	0	1	0.00	0.01	0.00	0.00	0.00	0.00	0.11	
cc	0	1	0.03	0.12	0.00	0.00	0.00	0.00	0.95	
alien	0	1	0.29	0.39	0.00	0.00	0.01	0.70	1.00	
succ	0	1	0.06	0.12	0.00	0.00	0.02	0.04	0.96	
thf	0	1	0.35	0.39	0.00	0.01	0.12	0.75	1.00	
wind	0	1	1.43	1.02	0.00	0.67	1.22	2.00	5.00	
fieldtime	0	1	7.32	0.82	4.42	6.83	7.21	7.74	14.54	
yday	0	1	0.43	0.02	0.35	0.41	0.42	0.45	0.64	
hr	0	1	0.30	0.03	0.18	0.28	0.30	0.32	0.61	

```r
round(cor(x[, sapply(x, is.numeric)]), 2)
```

```
               y  year longitude latitude   ahm   pet   ffp   map   mat  mcmt
y           1.00 -0.07     -0.26     0.46 -0.52 -0.50 -0.49  0.29 -0.46 -0.34
year       -0.07  1.00      0.17     0.01  0.13  0.06  0.09 -0.22 -0.03 -0.11
longitude  -0.26  0.17      1.00    -0.30  0.48  0.40  0.39 -0.52  0.20 -0.03
latitude    0.46  0.01     -0.30     1.00 -0.76 -0.87 -0.61  0.07 -0.96 -0.91
ahm        -0.52  0.13      0.48    -0.76  1.00  0.91  0.80 -0.63  0.78  0.53
pet        -0.50  0.06      0.40    -0.87  0.91  1.00  0.75 -0.44  0.90  0.68
```

ffp	-0.49	0.09	0.39	-0.61	0.80	0.75	1.00	-0.46	0.72	0.49
map	0.29	-0.22	-0.52	0.07	-0.63	-0.44	-0.46	1.00	-0.10	0.23
mat	-0.46	-0.03	0.20	-0.96	0.78	0.90	0.72	-0.10	1.00	0.92
mcmt	-0.34	-0.11	-0.03	-0.91	0.53	0.68	0.49	0.23	0.92	1.00
mwmt	-0.46	0.18	0.62	-0.48	0.88	0.79	0.81	-0.81	0.53	0.18
pforest	0.60	-0.18	-0.37	0.63	-0.71	-0.67	-0.63	0.42	-0.60	-0.45
pconif	0.59	-0.12	-0.19	0.49	-0.59	-0.61	-0.57	0.39	-0.52	-0.36
pwet	0.46	-0.07	-0.12	0.52	-0.45	-0.45	-0.44	0.10	-0.52	-0.49
poldfor	0.42	-0.17	-0.45	0.29	-0.48	-0.39	-0.44	0.49	-0.27	-0.10
cult	-0.49	0.15	0.20	-0.55	0.52	0.57	0.53	-0.23	0.57	0.47
urbind	-0.11	-0.03	0.00	-0.13	0.07	0.11	0.12	0.01	0.15	0.14
softlin	0.01	0.02	-0.15	-0.12	0.05	0.13	0.07	0.07	0.15	0.16
hardlin	-0.22	-0.05	0.02	-0.26	0.20	0.24	0.22	-0.03	0.28	0.26
cc	0.11	-0.14	-0.23	0.07	-0.19	-0.15	-0.10	0.26	-0.03	0.08
alien	-0.50	0.14	0.20	-0.57	0.52	0.58	0.54	-0.23	0.59	0.49
succ	0.11	-0.13	-0.26	0.04	-0.17	-0.11	-0.08	0.27	0.01	0.11
thf	-0.47	0.10	0.12	-0.56	0.47	0.55	0.52	-0.15	0.59	0.53
wind	-0.26	-0.03	0.10	-0.31	0.29	0.29	0.24	-0.12	0.30	0.27
fieldtime	0.19	-0.14	-0.25	0.10	-0.24	-0.20	-0.23	0.32	-0.09	0.02
yday	-0.03	-0.08	-0.03	0.07	-0.04	-0.09	0.04	0.04	-0.06	-0.04
hr	0.19	-0.14	-0.25	0.10	-0.24	-0.20	-0.23	0.32	-0.09	0.02

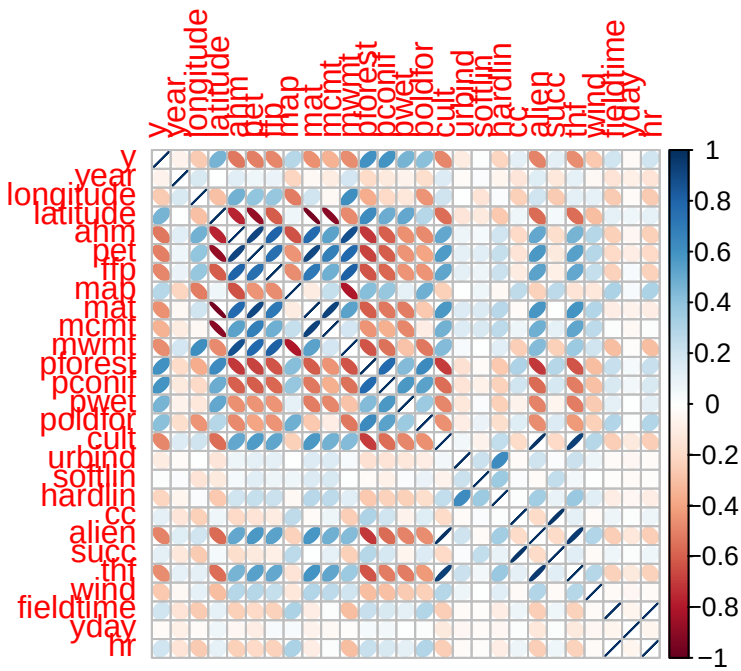
	mwmt	pforest	pconif	pwet	poldfor	cult	urbind	softlin	hardlin	cc
y	-0.46	0.60	0.59	0.46	0.42	-0.49	-0.11	0.01	-0.22	0.11
year	0.18	-0.18	-0.12	-0.07	-0.17	0.15	-0.03	0.02	-0.05	-0.14
longitude	0.62	-0.37	-0.19	-0.12	-0.45	0.20	0.00	-0.15	0.02	-0.23
latitude	-0.48	0.63	0.49	0.52	0.29	-0.55	-0.13	-0.12	-0.26	0.07
ahm	0.88	-0.71	-0.59	-0.45	-0.48	0.52	0.07	0.05	0.20	-0.19
pet	0.79	-0.67	-0.61	-0.45	-0.39	0.57	0.11	0.13	0.24	-0.15
ffp	0.81	-0.63	-0.57	-0.44	-0.44	0.53	0.12	0.07	0.22	-0.10
map	-0.81	0.42	0.39	0.10	0.49	-0.23	0.01	0.07	-0.03	0.26
mat	0.53	-0.60	-0.52	-0.52	-0.27	0.57	0.15	0.15	0.28	-0.03
mcmt	0.18	-0.45	-0.36	-0.49	-0.10	0.47	0.14	0.16	0.26	0.08
mwmt	1.00	-0.63	-0.55	-0.29	-0.53	0.43	0.04	0.01	0.12	-0.25
pforest	-0.63	1.00	0.78	0.43	0.64	-0.70	-0.16	-0.02	-0.27	0.30
pconif	-0.55	0.78	1.00	0.57	0.54	-0.55	-0.14	-0.08	-0.23	0.20
pwet	-0.29	0.43	0.57	1.00	0.37	-0.48	-0.13	0.03	-0.23	-

0.07										
poldfor	-0.53	0.64	0.54	0.37	1.00	-0.46	-0.10	0.09	-0.17	0.13
cult	0.43	-0.70	-0.55	-0.48	-0.46	1.00	0.05	-0.06	0.24	-
0.19										
urbind	0.04	-0.16	-0.14	-0.13	-0.10	0.05	1.00	0.22	0.62	-
0.03										
softlin	0.01	-0.02	-0.08	0.03	0.09	-0.06	0.22	1.00	0.36	0.02
hardlin	0.12	-0.27	-0.23	-0.23	-0.17	0.24	0.62	0.36	1.00	0.02
cc	-0.25	0.30	0.20	-0.07	0.13	-0.19	-0.03	0.02	0.02	1.00
alien	0.43	-0.71	-0.56	-0.50	-0.46	0.99	0.20	-0.02	0.35	-
0.18										
succ	-0.24	0.29	0.17	-0.06	0.15	-0.19	0.02	0.24	0.10	0.97
thf	0.36	-0.62	-0.52	-0.52	-0.42	0.94	0.21	0.06	0.38	0.12
wind	0.19	-0.31	-0.29	-0.26	-0.24	0.28	0.08	0.00	0.12	-
0.02										
fieldtime	-0.30	0.22	0.17	0.12	0.30	-0.24	-0.02	-0.02	-0.09	0.07
yday	-0.01	0.05	0.05	0.01	0.02	-0.12	-0.05	-0.02	-0.04	0.02
hr	-0.30	0.22	0.17	0.12	0.30	-0.24	-0.02	-0.02	-0.09	0.07

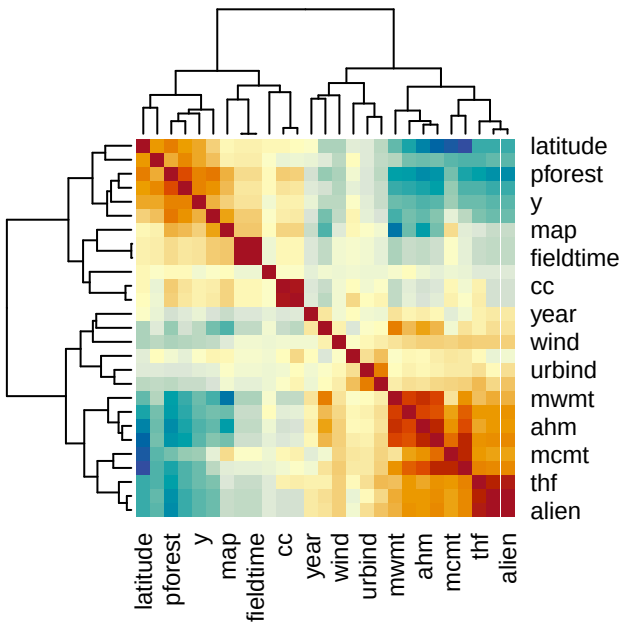
	alien	succ	thf	wind	fieldtime	yday	hr
y	-0.50	0.11	-0.47	-0.26	0.19	-0.03	0.19
year	0.14	-0.13	0.10	-0.03	-0.14	-0.08	-0.14
longitude	0.20	-0.26	0.12	0.10	-0.25	-0.03	-0.25
latitude	-0.57	0.04	-0.56	-0.31	0.10	0.07	0.10
ahm	0.52	-0.17	0.47	0.29	-0.24	-0.04	-0.24
pet	0.58	-0.11	0.55	0.29	-0.20	-0.09	-0.20
ffp	0.54	-0.08	0.52	0.24	-0.23	0.04	-0.23
map	-0.23	0.27	-0.15	-0.12	0.32	0.04	0.32
mat	0.59	0.01	0.59	0.30	-0.09	-0.06	-0.09
mcmt	0.49	0.11	0.53	0.27	0.02	-0.04	0.02
mwmt	0.43	-0.24	0.36	0.19	-0.30	-0.01	-0.30
pforest	-0.71	0.29	-0.62	-0.31	0.22	0.05	0.22
pconif	-0.56	0.17	-0.52	-0.29	0.17	0.05	0.17
pwet	-0.50	-0.06	-0.52	-0.26	0.12	0.01	0.12
poldfor	-0.46	0.15	-0.42	-0.24	0.30	0.02	0.30
cult	0.99	-0.19	0.94	0.28	-0.24	-0.12	-0.24
urbind	0.20	0.02	0.21	0.08	-0.02	-0.05	-0.02
softlin	-0.02	0.24	0.06	0.00	-0.02	-0.02	-0.02
hardlin	0.35	0.10	0.38	0.12	-0.09	-0.04	-0.09
cc	-0.18	0.97	0.12	-0.02	0.07	0.02	0.07
alien	1.00	-0.18	0.95	0.29	-0.24	-0.12	-0.24
succ	-0.18	1.00	0.13	-0.02	0.07	0.01	0.07
thf	0.95	0.13	1.00	0.29	-0.22	-0.12	-0.22
wind	0.29	-0.02	0.29	1.00	-0.03	-0.04	-0.03

```
fieldtime -0.24  0.07 -0.22 -0.03      1.00 -0.05  1.00
yday      -0.12  0.01 -0.12 -0.04     -0.05  1.00 -0.05
hr        -0.24  0.07 -0.22 -0.03      1.00 -0.05  1.00
```

```
corrplot::corrplot(cor(x[, sapply(x, is.numeric)]), "ellipse")
```




```
heatmap(cor(x[, sapply(x, is.numeric)]),
    symm = TRUE,
    col = hcl.colors(100, "RdYlBu", rev = TRUE)
)
```

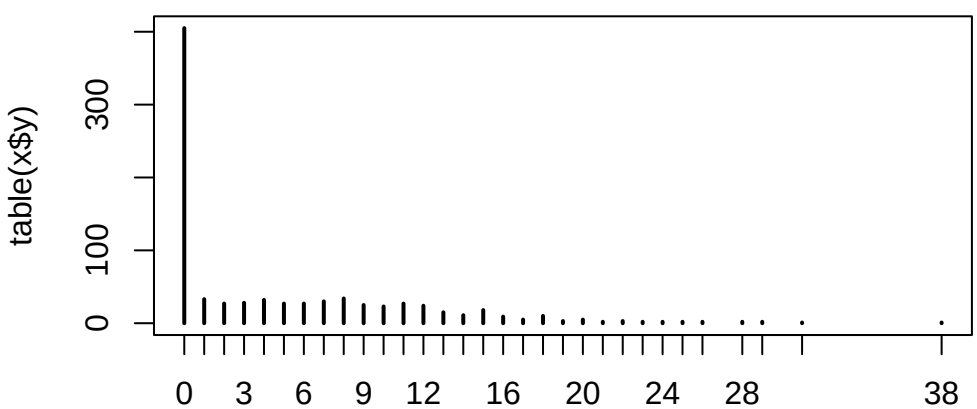


Explore the response variable:

```
table(x$y)
```

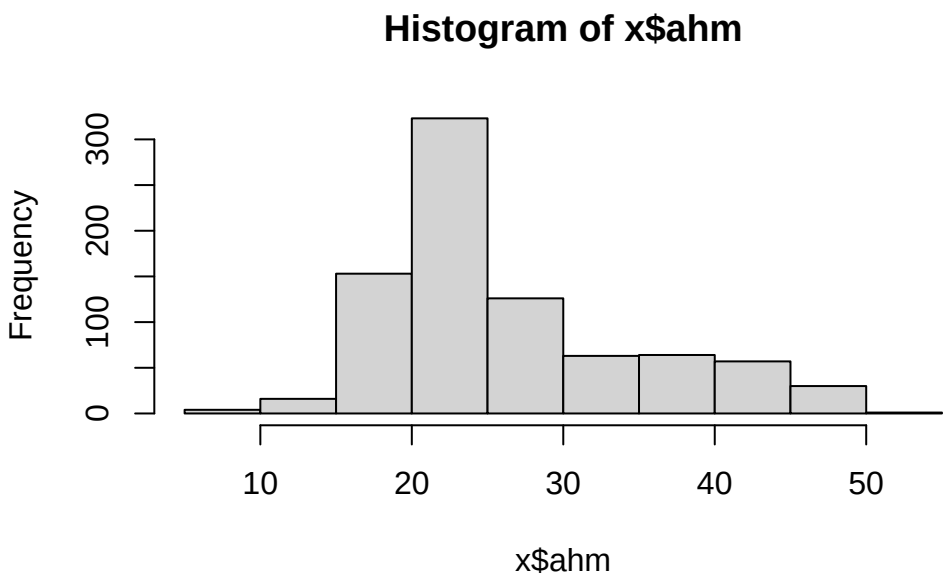
```
  0   1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19
405  33  27  28  32  27  27  30  34  25  23  27  24  15  11  18   9   5  10   3
 20  21  22  23  24  25  26  28  29  31  38
  5   2   3   2   2   2   2   2   2   1   1
```

```
plot(table(x$y))
```

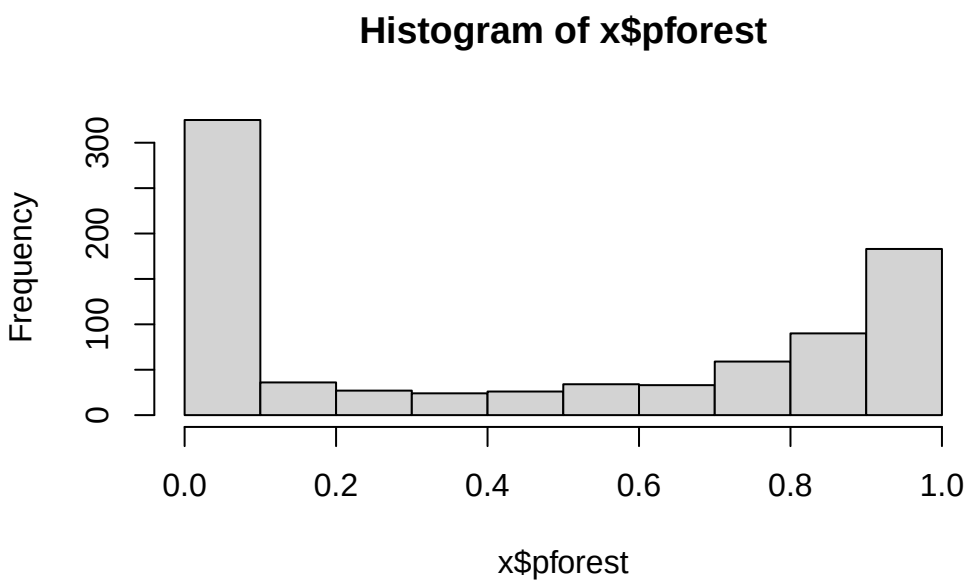


Explore the predictor variables:

```r
hist(x$ahm)
```




```r
hist(x$pforest)
```



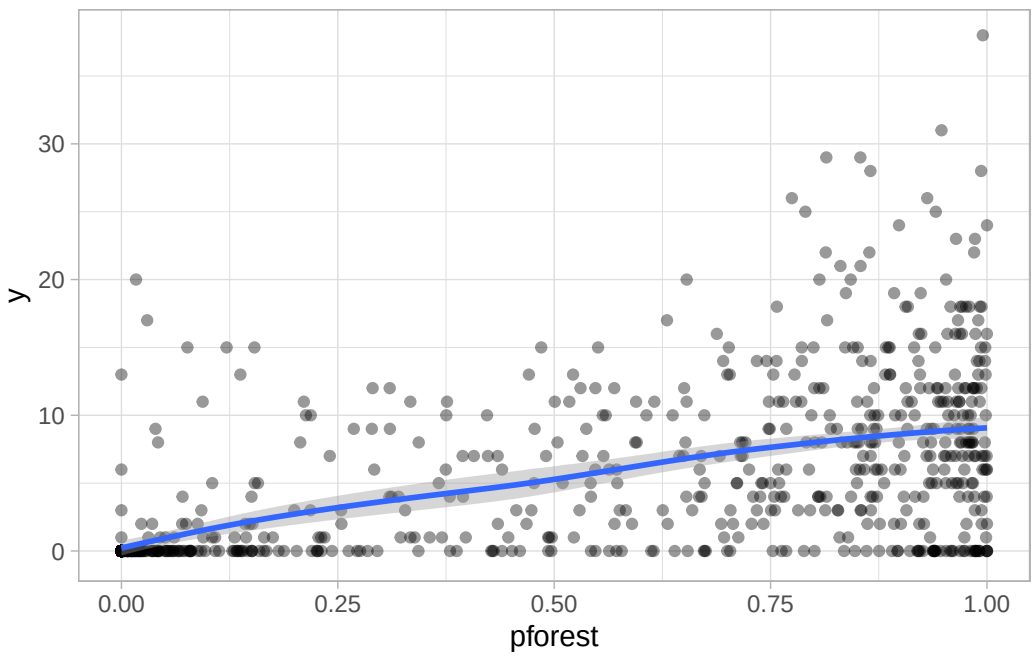
Bivariate relationships:

```r
# where to begin
cor(x[, sapply(x, is.numeric)])[, "y"] |> sort()
```

```
        ahm         pet       alien        cult         ffp         thf 
-0.52146866 -0.50260785 -0.49661534 -0.48856911 -0.48538551 -0.46589038 
       mwmt         mat        mcmt        wind   longitude     hardlin 
-0.46306993 -0.45975887 -0.34431295 -0.26272086 -0.25687592 -0.21936446 
     urbind        year        yday     softlin        succ          cc 
-0.10648235 -0.06654048 -0.02816788  0.01415377  0.11299109  0.11326442 
  fieldtime          hr         map     poldfor    latitude        pwet 
 0.19288915  0.19288915  0.28501472  0.41909313  0.46090872  0.46183944 
     pconif     pforest           y 
 0.59250179  0.60268088  1.00000000 
```

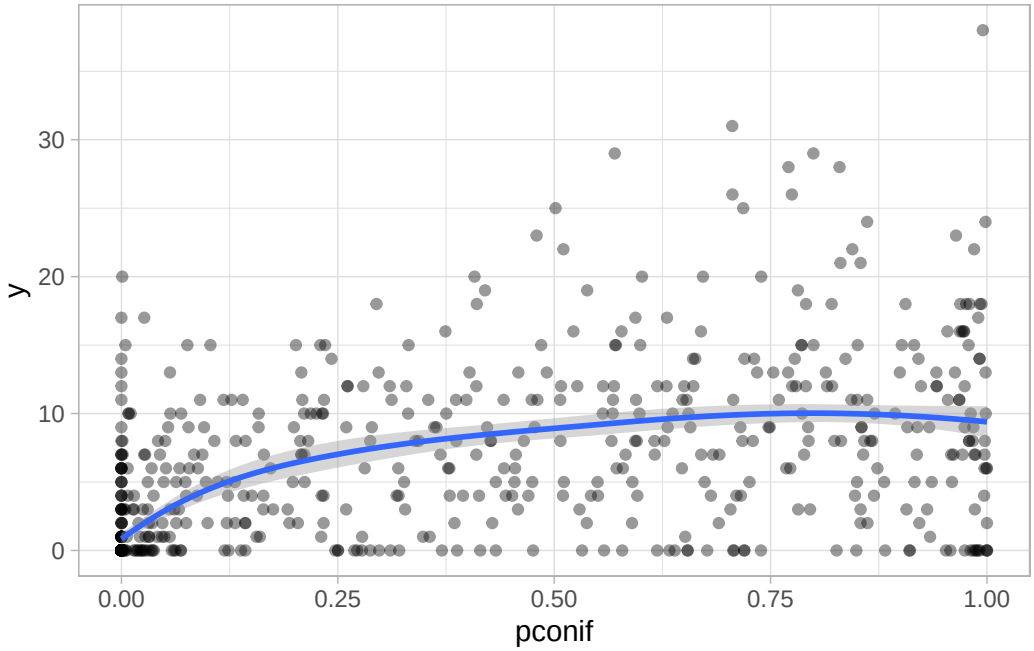
```
x |>
    ggplot(aes(x = pforest, y = y)) +
    geom_point(alpha = 0.4) +
    geom_smooth(method = "loess", se = TRUE)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

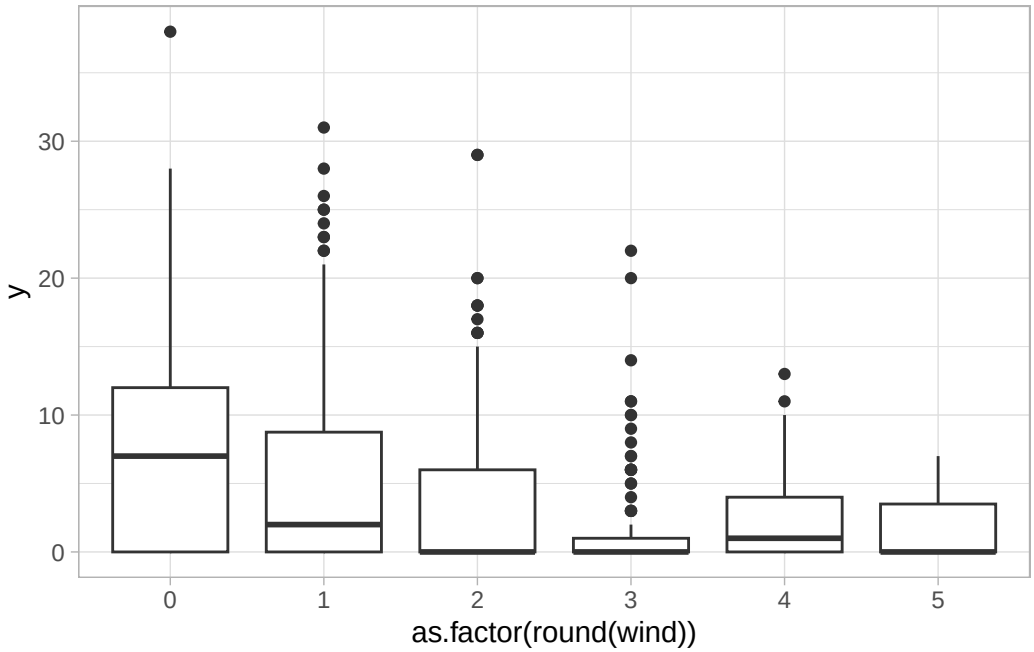



```
x |>
    ggplot(aes(x = pconif, y = y)) +
    geom_point(alpha = 0.4) +
    geom_smooth(method = "loess", se = TRUE)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

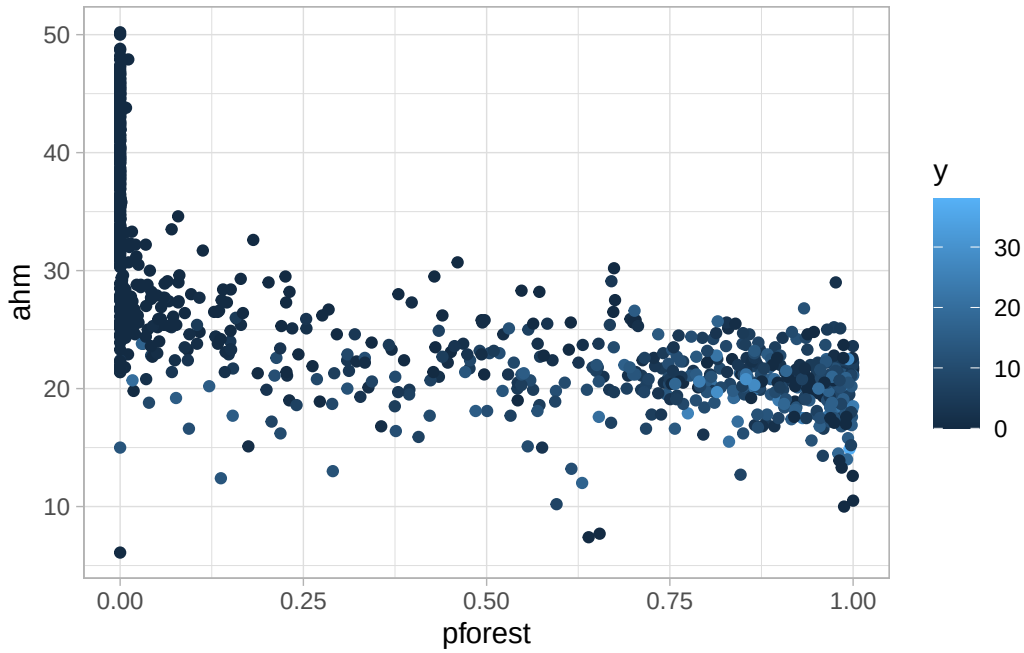



```r
x |>
    ggplot(aes(x = as.factor(round(wind)), y = y)) +
    geom_boxplot()
```

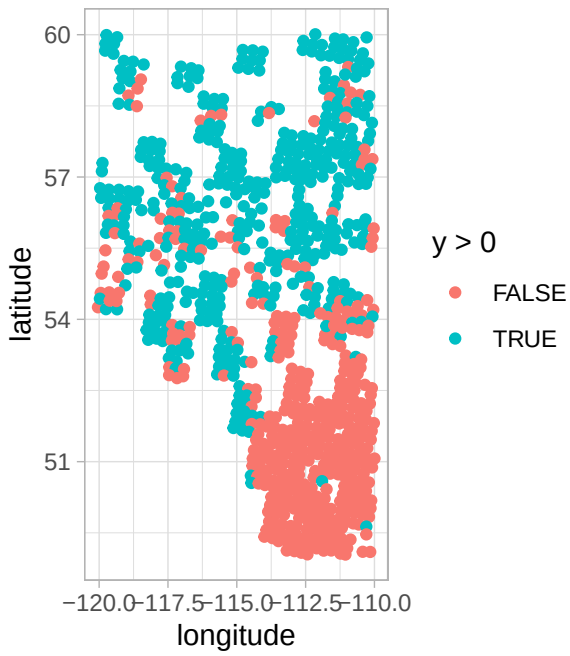


Trivariate relationships:

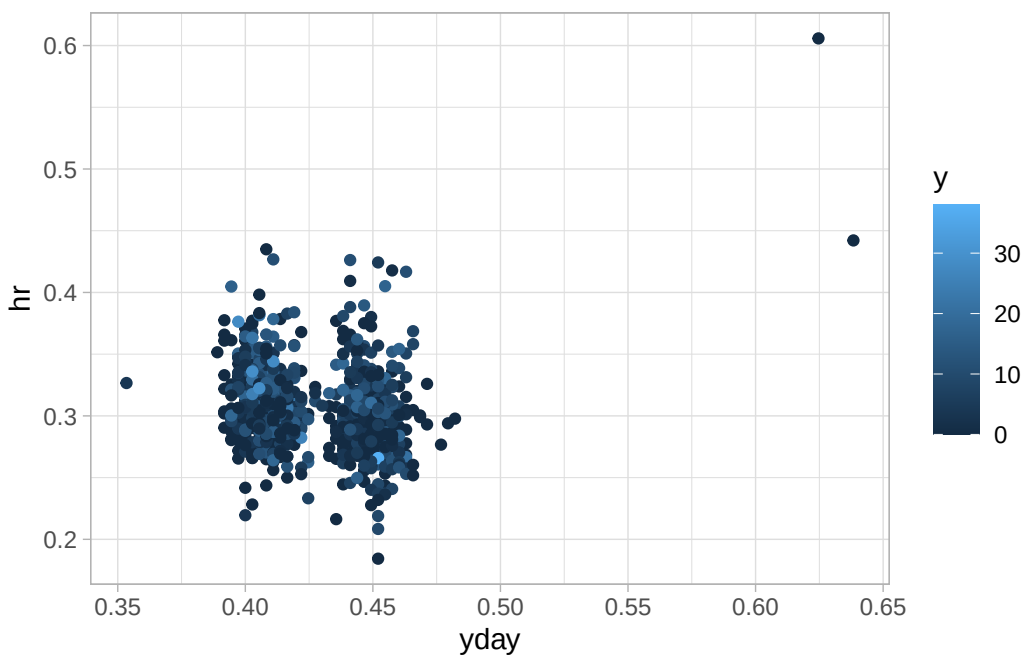
```
x |>
    ggplot(aes(x = pforest, y = ahm, col = y)) +
    geom_point()
```



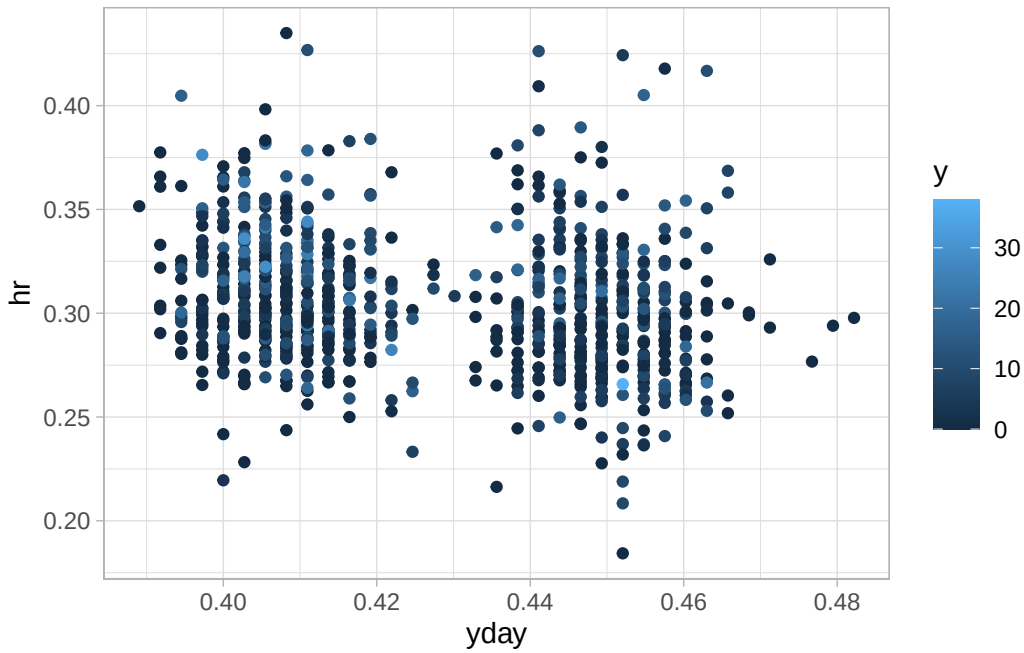

```
x |>
    ggplot(aes(x = longitude, y = latitude)) +
    geom_point(aes(color = y > 0)) +
    coord_quickmap()
```




```
x |>
    ggplot(aes(x = yday, y = hr, col = y)) +
    geom_point()
```



```r
# need to filter some extreme values for better visualization
x <- x |>
    dplyr::filter(
        hr > 0.1,
        hr < 0.6,
        yday > 0.37,
        yday < 0.5
    )
x |>
    ggplot(aes(x = yday, y = hr, col = y)) +
    geom_point()
```

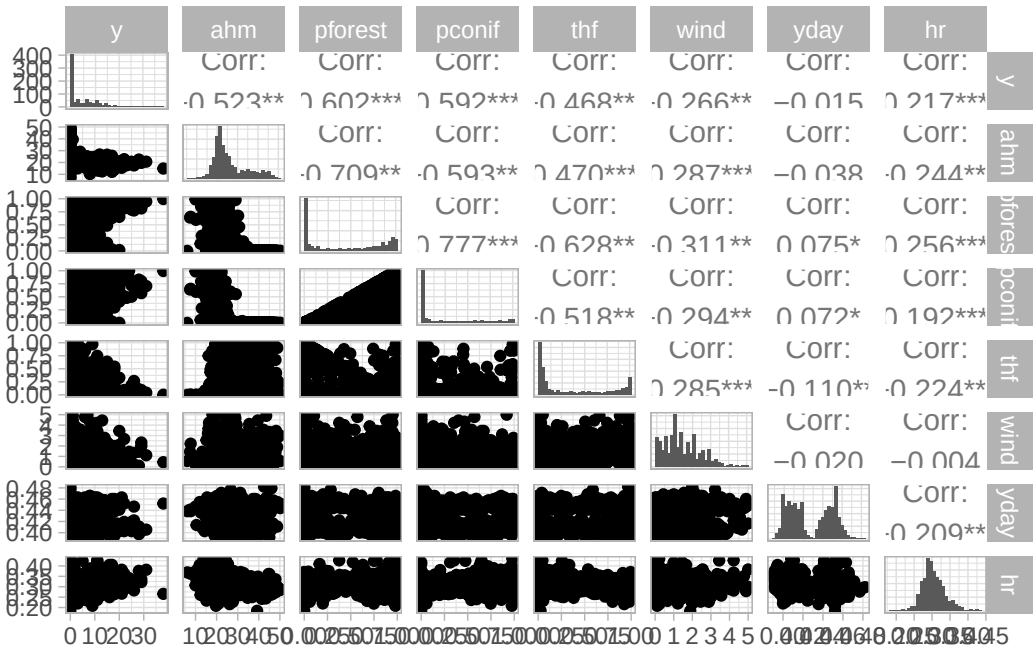


Multivariate relationships:

```r
GGally::ggpairs(
    x,
    columns = c("y", "ahm", "pforest", "pconif", "thf", "wind", "yday", "hr"),
    diag = list(continuous = GGally::wrap("barDiag"))
)
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



JOSM data

Covariates

Variables that co-vary with the response variable. Also called as independent variables, predictors.

Let's continue with the JOSM data set:

```
x <- detect::josm$surveys |>
    dplyr::select(
        Longitude,
        Latitude,
        WindStart,
        TSSR,
        DAY,
```

```
        Open,
        Water,
        Decid,
        OpenWet,
        Conif,
        ConifWet,
        Agr,
        UrbInd,
        SoftLin,
        Roads
    )
```

STOP AND EXPLAIN EACH VARIABLE!

Variable types

What type of variables we have? You can use the `str()` function to reveal the structure of R objects:

```
str(x)
```

```
'data.frame':   4569 obs. of  15 variables:
 $ Longitude: num  -113 -113 -113 -113 -113 ...
 $ Latitude : num  55.2 55.2 55.2 55.2 55.2 ...
 $ WindStart: int  0 0 0 0 0 1 2 0 0 0 ...
 $ TSSR     : num  0.0132 0.0666 0.0125 0.041 0.1263 ...
 $ DAY      : num  0.471 0.471 0.471 0.471 0.471 ...
 $ Open     : num  0 0 0 0 0 0 0 0 0 0 ...
 $ Water    : num  0.02055 0.00752 0.00752 0.03284 0.00416 ...
 $ Decid    : num  0.8569 0.0443 0.0443 0.8823 0.8139 ...
 $ OpenWet  : num  0.00315 0.58355 0.58355 0.06961 0.11164 ...
 $ Conif    : num  0.0249 0 0 0 0 ...
 $ ConifWet : num  0.0368 0.3456 0.3456 0 0.0018 ...
 $ Agr      : num  0 0 0 0 0 0 0 0 0 0 ...
 $ UrbInd   : num  0 0 0 0.01056 0.00839 ...
 $ SoftLin  : num  0.00521 0.01811 0.01811 0.00141 0.01349 ...
 $ Roads    : num  0.052461 0.000913 0.000913 0.003307 0.046578 ...
```

We see that these variables are all continuous.

However, `WindStart` has very few distinct values, so we could treat it as ordinal (ordered factor):

```r
table(x$WindStart)
```

```
   0    1    2    3    4    5    6 
2471  926  844  287   31    5    3 
```

```r
x$WindOrd <- as.ordered(x$WindStart)
str(x$WindOrd)
```

```
 Ord.factor w/ 7 levels "0"<"1"<"2"<"3"<..: 1 1 1 1 1 2 3 1 1 1 ...
```

```r
levels(x$WindOrd)
```

```
[1] "0" "1" "2" "3" "4" "5" "6"
```

Sometimes variables are binary. In R these can be logical (`TRUE`/`FALSE`) or coded as 0/1. Often we make such variables by discretizing other continuous or ordinal variables. E.g. We can create a binary wind variable:

```r
x$Wind01 <- ifelse(x$WindStart > 0, 1, 0)
table(x$WindStart, x$Wind01)
```

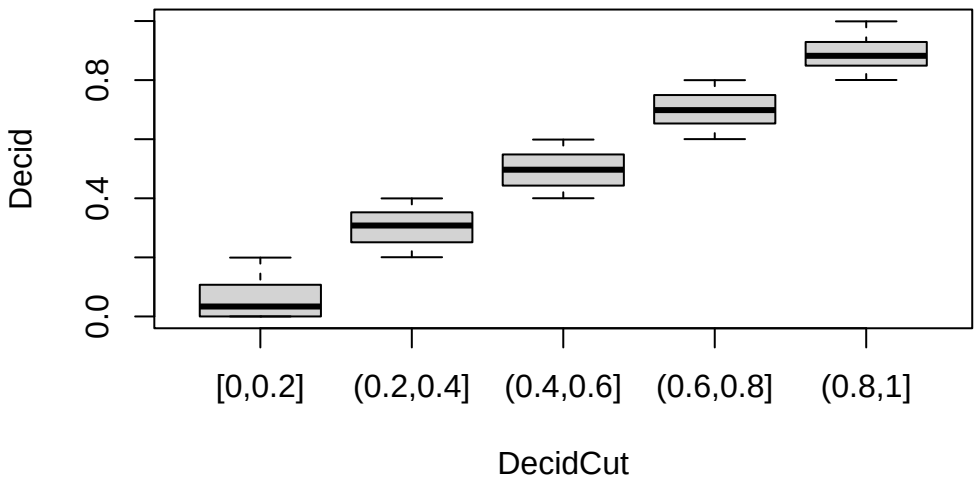
```
       0    1
  0 2471    0
  1    0  926
  2    0  844
  3    0  287
  4    0   31
  5    0    5
  6    0    3
```

Some categorical variables depend on some kind of classification, for example we can cut a continuous variable into bins:

```r
x$DecidCut <- cut(x$Decid, seq(0, 1, 0.2), include.lowest = TRUE)
table(x$DecidCut)
```

```
  [0,0.2] (0.2,0.4] (0.4,0.6] (0.6,0.8]   (0.8,1]
     1592       878       857       638       604
```

```
boxplot(Decid ~ DecidCut, x)
```



We can also inspect the land cover proportions that add up to 1 for each row.

```
## define column names
cn <- c(
    "Open", "Water", "Agr", "UrbInd", "SoftLin", "Roads", "Decid",
    "OpenWet", "Conif", "ConifWet"
)
## these sum to 1
summary(rowSums(x[, cn]))
```

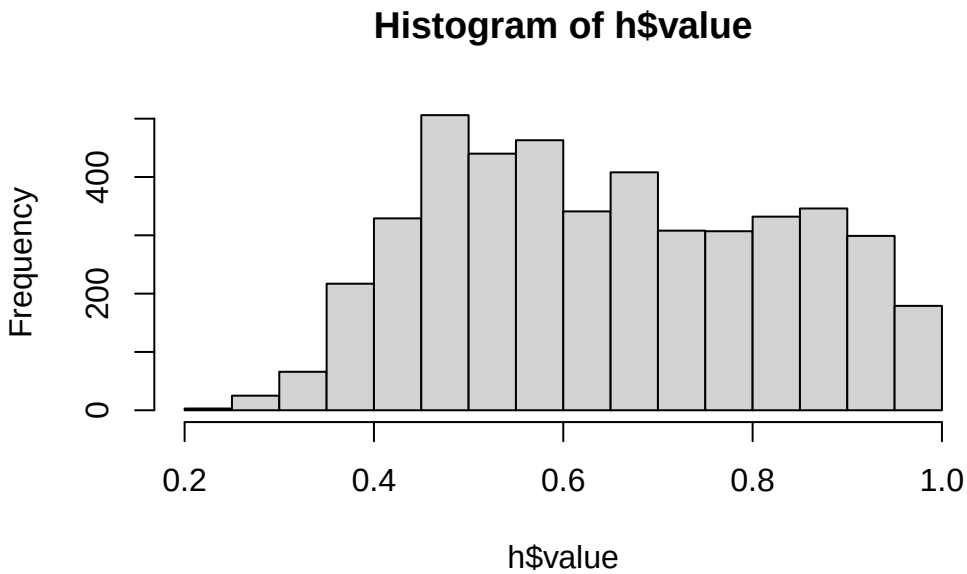
```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1       1       1       1       1       1
```

The `find_max()` function finds the maximum value in each row, the output contains the value and the column where it was found, we can turn that into the dominant land cover type encoded in `HAB`:

```
h <- mefa4::find_max(x[, cn])
head(h)
```

```
          index     value
CL10102   Decid 0.8569106
CL10106 OpenWet 0.5835472
CL10108 OpenWet 0.5835472
CL10109   Decid 0.8822829
CL10111   Decid 0.8139365
CL10112   Decid 0.8139365
```

```r
hist(h$value)
```




```r
table(h$index)
```

```
    Open    Water      Agr   UrbInd  SoftLin    Roads    Decid  OpenWet 
      12       10        4       14        0        2     2084      160 
   Conif ConifWet 
     745     1538 
```

```r
x$HAB <- droplevels(h$index) # drop empty levels
x$DEC <- ifelse(x$HAB == "Decid", 1, 0)
```

Other types of categorical variables are truly discrete, like observer, where there are no underlying continuous data:

```r
table(detect::josm$surveys$ObserverID)
```

```
  2  14  15  19  22  26  27  28  32  41  48  57  59  63  64  65  69  75  82  86
186 223  88 267   7 191 307 227 231 140 170 126 222 240 127 154 302 127 173 280
 87  89  93  98
166  99 244 272
```

When the number of categories increase and approach the sample size, we can consider treating these variables as random effects. E.g. the `SurveyArea` variable that has 271 levels.

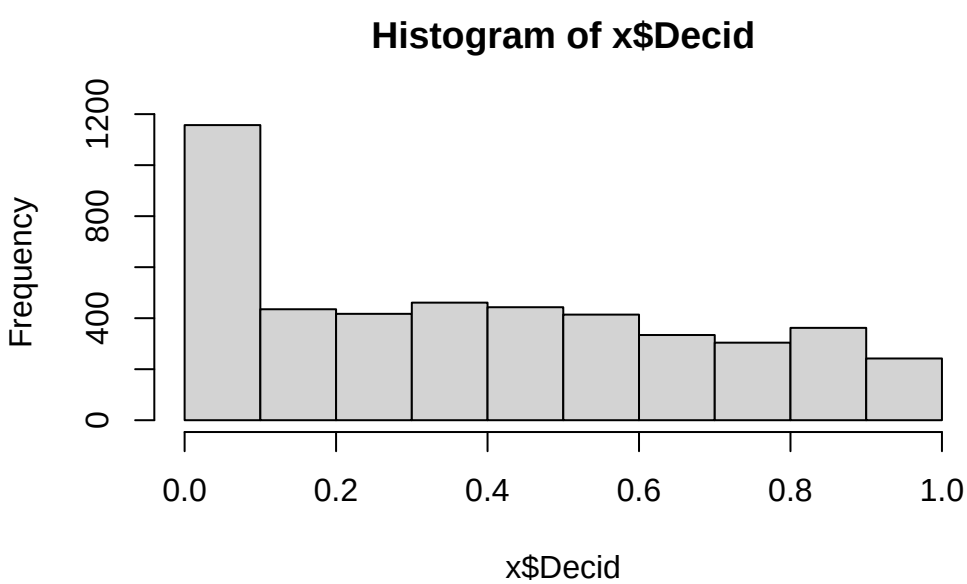
Data exploration

We should inspect each variable that we want to use as a covariate. Here are some of the most important functions:

```r
summary(x$Decid) # check mean, range, missing values
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
0.00000 0.09852 0.36132 0.38717 0.63232 0.99891 
```

```r
hist(x$Decid) # check skew and outliers
```



A nice way of getting all of the above nicely formatted is to use the skimr package:

```r
skimr::skim(x)
```

Table 9: Data summary

Name	x
Number of rows	4569
Number of columns	20
Column type frequency:	
factor	3
numeric	17
Group variables	None

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
WindOrd	2	1	TRUE	7	0: 2471, 1: 926, 2: 844, 3: 287
DecidCut	0	1	FALSE	5	[0,: 1592, (0.: 878, (0.: 857, (0.: 638
HAB	0	1	FALSE	9	Dec: 2084, Con: 1538, Con: 745, Ope: 160

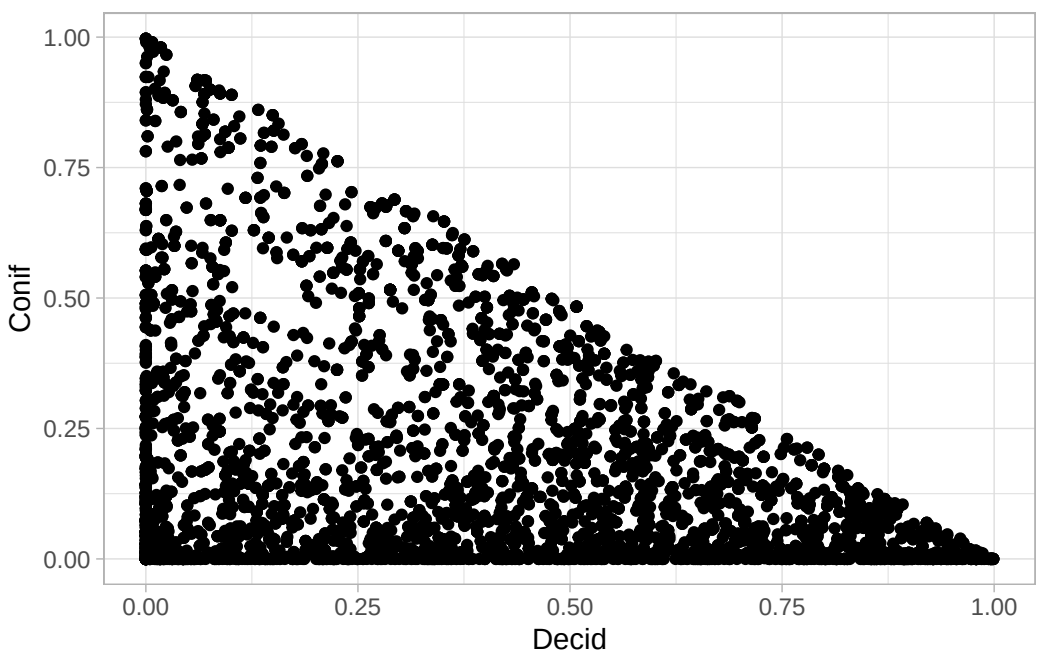
Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
Longitude	0	1	-113.20	1.96	-117.09	-114.55	-112.75	-111.49	-110.06	
Latitude	0	1	56.20	0.74	54.58	55.58	56.15	56.81	57.71	
WindStart	2	1	0.80	1.02	0.00	0.00	0.00	2.00	6.00	
TSSR	0	1	0.10	0.06	-0.03	0.05	0.10	0.16	0.24	
DAY	0	1	0.45	0.03	0.39	0.42	0.45	0.47	0.50	
Open	0	1	0.01	0.04	0.00	0.00	0.00	0.00	0.65	
Water	0	1	0.01	0.04	0.00	0.00	0.00	0.00	0.81	
Decid	0	1	0.39	0.30	0.00	0.10	0.36	0.63	1.00	
OpenWet	0	1	0.07	0.12	0.00	0.00	0.02	0.08	0.87	

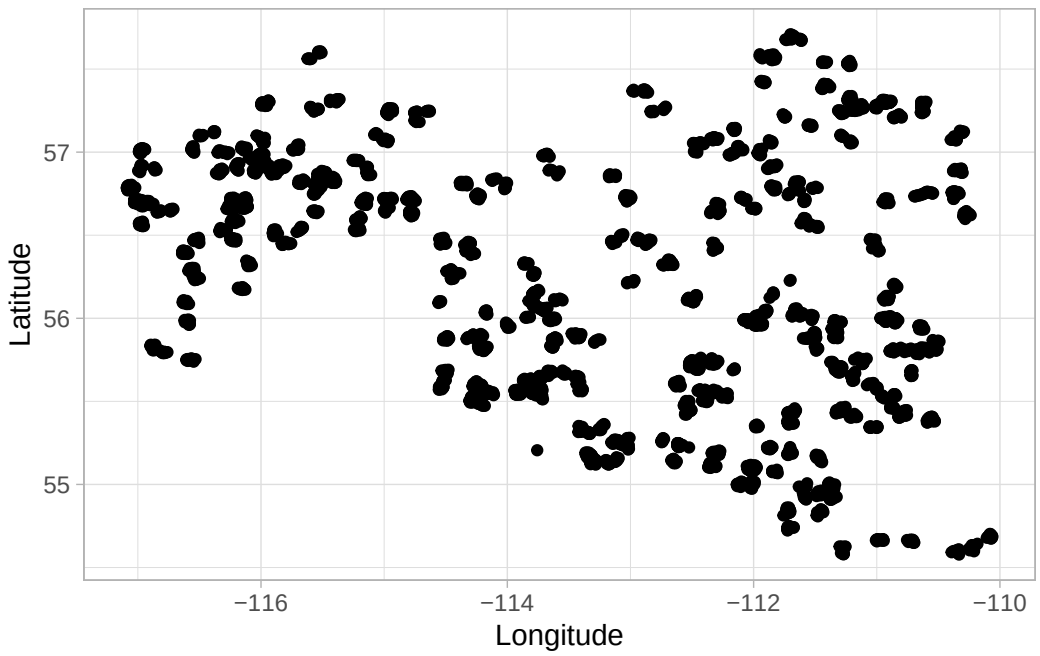
skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
Conif	0	1	0.18	0.22	0.00	0.01	0.08	0.27	1.00	
ConifWet	0	1	0.30	0.29	0.00	0.04	0.20	0.50	1.00	
Agr	0	1	0.00	0.02	0.00	0.00	0.00	0.00	0.79	
UrbInd	0	1	0.01	0.05	0.00	0.00	0.00	0.01	1.00	
SoftLin	0	1	0.02	0.02	0.00	0.01	0.01	0.03	0.25	
Roads	0	1	0.01	0.02	0.00	0.00	0.00	0.01	0.27	
Wind01	2	1	0.46	0.50	0.00	0.00	0.00	1.00	1.00	
DEC	0	1	0.46	0.50	0.00	0.00	0.00	1.00	1.00	

To explore relationships between variables, make scatter and box plots:

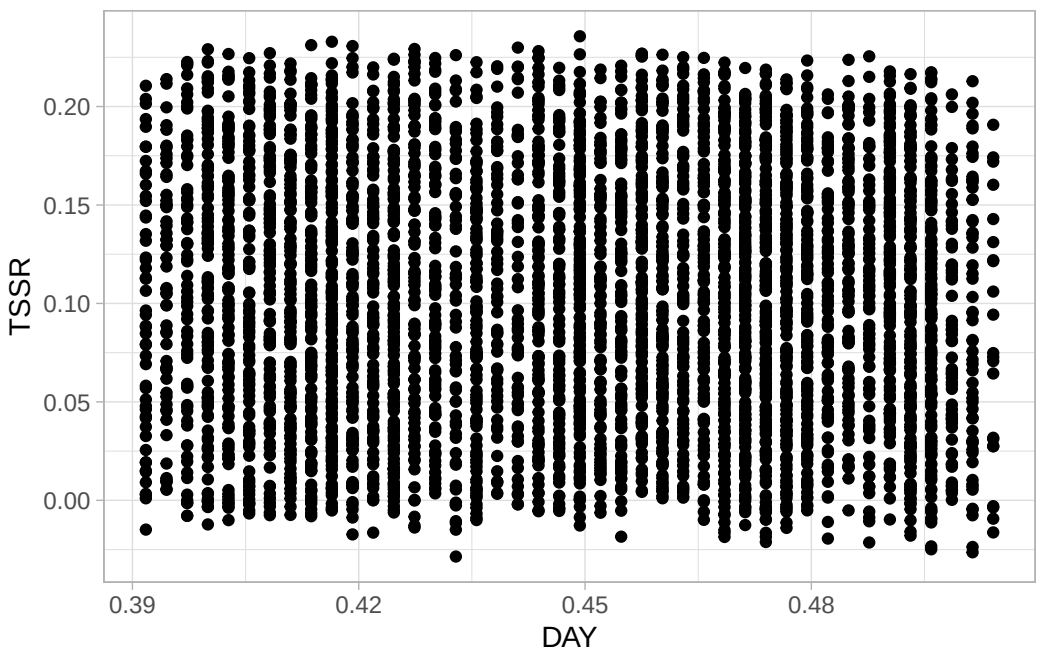
```
x |> ggplot(aes(x = Decid, y = Conif)) +
    geom_point()
```



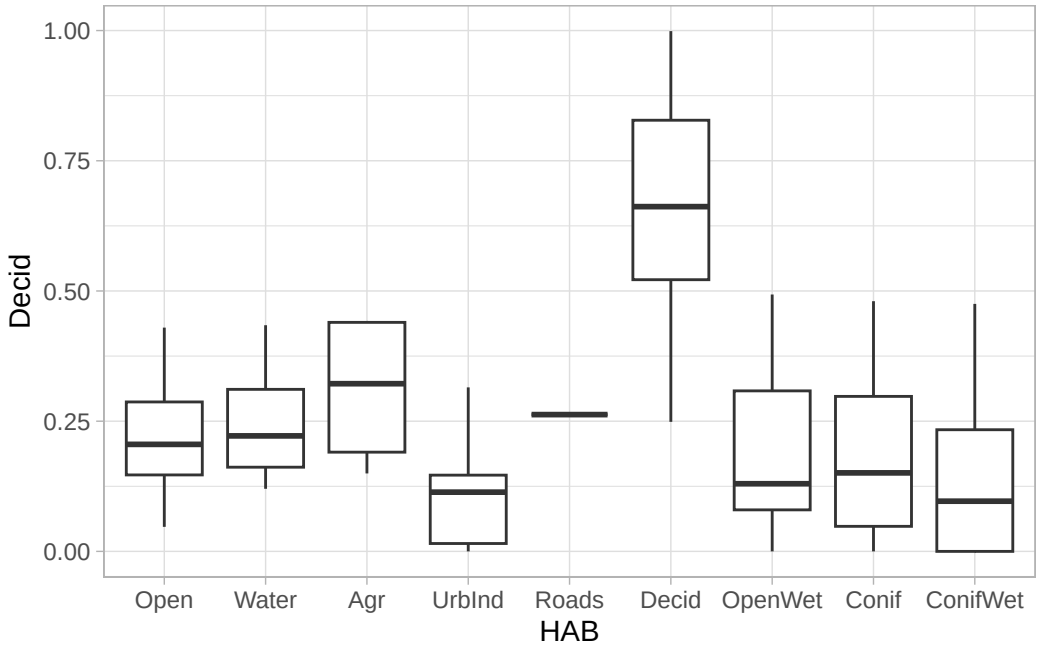

```
x |> ggplot(aes(x = Longitude, y = Latitude)) +
    geom_point()
```




```
x |> ggplot(aes(x = DAY, y = TSSR)) +
    geom_point()
```

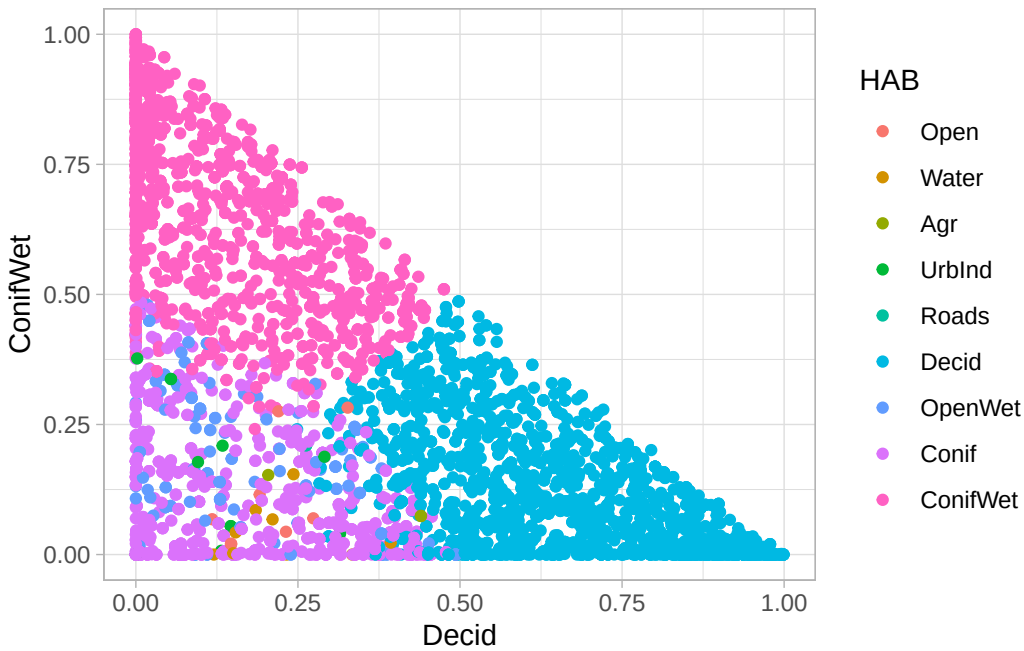


```
x |> ggplot(aes(x = HAB, y = Decid)) +
    geom_boxplot()
```



We can present 3 variables as color scatter or bubble plots:

```
x |> ggplot(aes(x = Decid, y = ConifWet, col = HAB)) +
    geom_point()
```



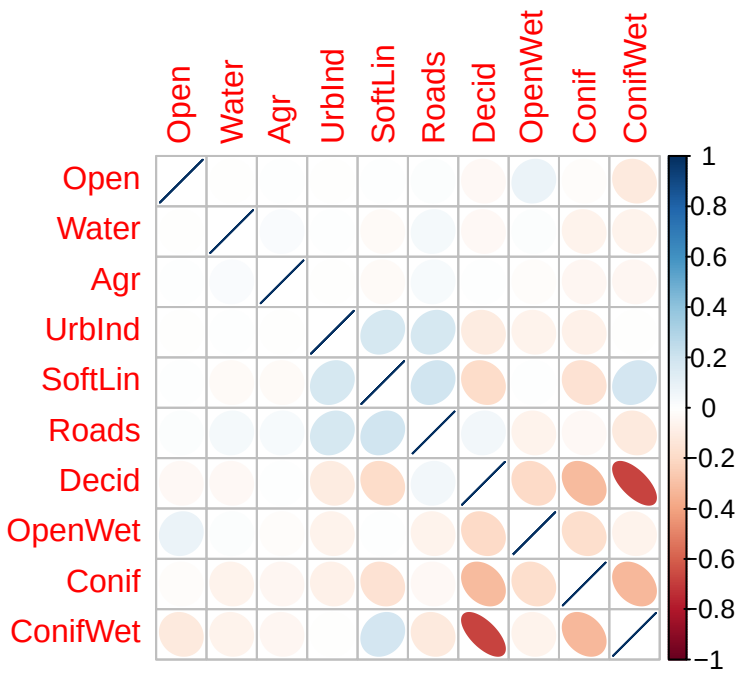
Multivariate exploration include checking correlations:

```
round(cor(x[, cn]), 3)
```

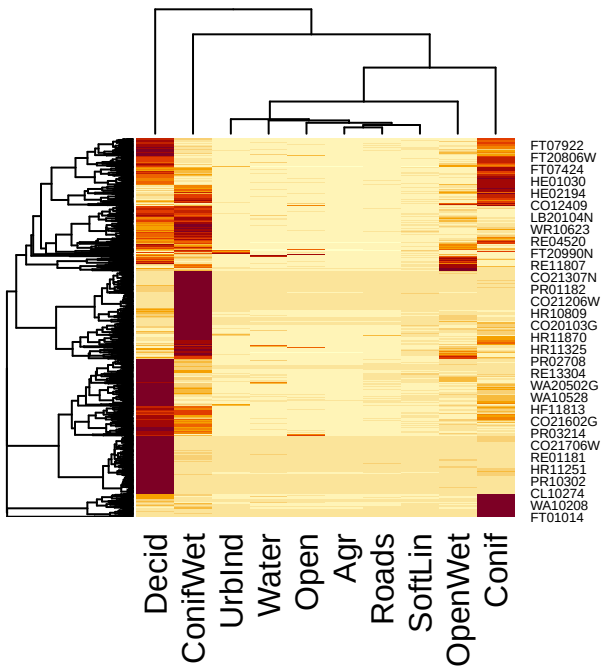
```
           Open  Water    Agr UrbInd SoftLin  Roads  Decid OpenWet  Conif
Open      1.000 -0.002  0.004 -0.007   0.006  0.011 -0.038   0.081 -0.019
Water    -0.002  1.000  0.023  0.007  -0.026  0.048 -0.039   0.012 -0.063
Agr       0.004  0.023  1.000 -0.004  -0.027  0.036  0.008  -0.012 -0.040
UrbInd   -0.007  0.007 -0.004  1.000   0.178  0.174 -0.100  -0.060 -0.070
SoftLin   0.006 -0.026 -0.027  0.178   1.000  0.193 -0.185   0.001 -0.152
Roads     0.011  0.048  0.036  0.174   0.193  1.000  0.055  -0.063 -0.037
Decid    -0.038 -0.039  0.008 -0.100  -0.185  0.055  1.000  -0.197 -0.314
OpenWet   0.081  0.012 -0.012 -0.060   0.001 -0.063 -0.197   1.000 -0.173
Conif    -0.019 -0.063 -0.040 -0.070  -0.152 -0.037 -0.314  -0.173  1.000
ConifWet -0.115 -0.068 -0.048 -0.008   0.187 -0.120 -0.674  -0.069 -0.330
         ConifWet
Open       -0.115
Water      -0.068
Agr        -0.048
UrbInd     -0.008
SoftLin     0.187
Roads      -0.120
Decid      -0.674
OpenWet    -0.069
```

```
Conif       -0.330
ConifWet     1.000
```

```
corrplot::corrplot(cor(x[, cn]), "ellipse")
```




```
heatmap(as.matrix(x[, cn]))
```



Variable transformations

We have seen examples of these:

- indicator variables (0/1)
- discretization (cut)

Other transformations include:

- sqrt: to tame outliers (not suitable for negative values)
- log: we'll see many use cases later
- polynomials: nonlinear terms (`x^2`, `x^3`, etc.)
- centering: keeps the distribution but shifts the mean
- scaling: keeps the distribution but shifts the range (often used with centering)

Compound variables

We can reduce correlation by combining variables together when those are additive:

```
x$FOR <- x$Decid + x$Conif + x$ConifWet
x$HF <- x$Agr + x$UrbInd + x$Roads + x$SoftLin
x$WET <- x$OpenWet + x$ConifWet + x$Water
```

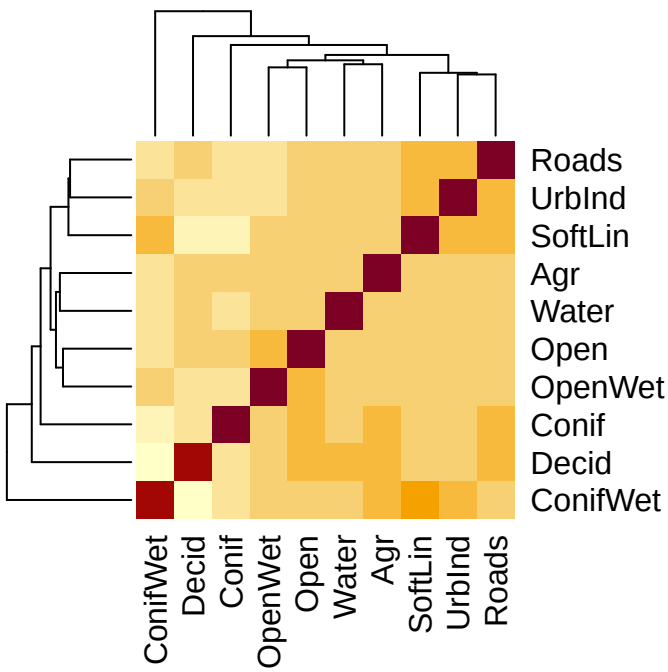
We can reduce colinearity by calculating variables relative to each other, like proportions or ratios (watch out for division by 0):

```
x$pDecid <- ifelse(x$FOR > 0, x$Decid / x$FOR, 0)
cor(x[, c("FOR", "pDecid")])
```

```
              FOR     pDecid
FOR    1.00000000 0.01124794
pDecid 0.01124794 1.00000000
```

We can also merge categories, e.g. based on their similarity:

```
heatmap(cor(x[, cn]))
```



Missing data

Let's check for missing values:

```
data.frame(missing = colSums(is.na(x))) |> filter(missing > 0)
```

```
          missing
WindStart       2
WindOrd         2
Wind01          2
```

There are only 2 rows with missing values, we could use `na.omit(x)` to drop these, or we can impute the values:

- pick a value randomly
- use the mean
- use the most common value (mode)
- Use other variables to predict the possible value

After inspecting the full data set, it turns out that site FT21204W experiences wind according to `WindEnd`:

```
sites <- rownames(x)[is.na(x$WindStart)]
detect::josm$surveys[detect::josm$surveys$SiteID %in% c("FT07424", "FT21204W"), ]
```

```
           SiteID SurveyArea Longitude Latitude       Date  StationID
FT07424   FT07424      FT074 -111.7273 55.18270 2012-06-28  FT07424-1
FT21204W FT21204W      FT212 -111.5415 57.15522 2014-06-30 FT21204W-1
         ObserverID  TimeStart VisitID WindStart PrecipStart TempStart
FT07424          57 8:20:00 AM       1        NA           0        NA
FT21204W         32 8:30:00 AM       1        NA          NA      20.3
         CloudStart WindEnd PrecipEnd TempEnd CloudEnd    TimeFin Noise
FT07424           0      NA         0      NA        0 8:30:00 AM     0
FT21204W         NA       3         0    20.3        0 8:40:00 AM     1
         OvernightRain            DateTime         SunRiseTime SunRiseFrac
FT07424           TRUE 2012-06-28 08:20:00 2012-06-28 04:49:30   0.2010437
FT21204W         FALSE 2014-06-30 08:30:00 2014-06-30 04:34:45   0.1907999
              TSSR OrdinalDay       DAY Open        Water Agr      UrbInd
FT07424  0.1461785        179 0.4904110    0 0.0023198943   0 0.006284186
FT21204W 0.1633668        180 0.4931507    0 0.0006266884   0 0.082233995
             SoftLin       Roads     Decid    OpenWet     Conif  ConifWet
FT07424  0.000000000 0.001462686 0.2005029 0.03073732 0.4908488 0.2678442
FT21204W 0.008925667 0.061308954 0.2785886 0.00000000 0.0000000 0.5683161
```

Let's set the value to 3 for site FT21204W, and the mode (0) for site FT07424:

```r
x[sites, "WindStart"] <- c(0, 3)
x[sites, "WindOrd"] <- c("0", "3")
x[sites, "Wind01"] <- c(0, 1)
```

Any more NAs left?

```r
any(is.na(x))
```

```
[1] FALSE
```

Response variable

We have manipulated the covariates and the species counts. Let's put them together in the same data frame.

```r
spp <- "OVEN" # change here if you want to use another species
detect::josm$species[spp, ]
```

```
     SpeciesID SpeciesName       ScientificName
OVEN      OVEN    Ovenbird Seiurus aurocapillus
```

```r
y <- mefa4::Xtab(~ SiteID + SpeciesID, detect::josm$counts)[, spp, drop = FALSE]
x$Count <- y[rownames(x), ]
```

Let's inspect the response variable:

```r
table(x$Count)
```

```
   0    1    2    3    4    5    6
2492  881  654  365  134   30   13
```

Question: How would we model the abundance of this species?

Working with species data

Data from Mahon et al. 2016 and Mahon et al. 2019.

Mahon, C. L., Holloway, G., Solymos, P., Cumming, S. G., Bayne, E. M., Schmiegelow, F. K. A., Song, S. J., 2016. Community structure and niche characteristics of upland and lowland western boreal birds at multiple spatial scales Forest Ecology and Management 361:99-116. https://doi.org/10.1016/j.foreco.2015.11.007

Mahon, C. L., Holloway, G., Bayne, E. M., Toms, J. D., 2019. Additive and interactive cumulative effects on boreal landbirds: winners and losers in a multi-stressor landscape Ecological Applications 29:e01895. https://doi.org/10.1002/eap.1895

```
d <- detect::josm$counts |>
    mutate(Behav = DetectType1, Count = 1) |>
    select(SiteID, SpeciesID, Count, Behav, Dur, Dis) |>
    arrange(SiteID, SpeciesID, Dur, Dis)
rownames(d) <- NULL
```

Point count data is usually organized in long format with the following information:

- the ID of the site or survey visit (we use `SiteID` because only 1 station and visit was done at each site)
- the species code (4 letters)
- the count (in this case each row is a distinct individual)
- the behavior code (C=call, S=song, V=visual)
- the time interval when the individual was first detected
- the distance interval in which the individual was first detected

```
summary(d)
```

```
     SiteID        SpeciesID         Count   Behav          Dur
 CO10712:  206   TEWA    : 5635   Min.   :1   C: 9180   0-3min :33648
 CO12627:  107   YRWA    : 4134   1st Qu.:1   S:41808   3-5min : 7220
 CO12724:   96   OVEN    : 4048   Median :1   V: 1384   5-10min:11504
 CO12780:   87   WTSP    : 3753   Mean   :1
 FT06718:   87   SWTH    : 3402   3rd Qu.:1
 CO12613:   85   CHSP    : 2096   Max.   :1
 (Other):51704   (Other):29304
      Dis
 0-50m  :23531
 50-100m:22230
 100+m  : 6611
```

```r
str(d)
```

```
'data.frame':   52372 obs. of  6 variables:
 $ SiteID   : Factor w/ 4569 levels "CL10102","CL10106",..: 1 1 1 1 1 1 1 1 2 2 ...
 $ SpeciesID: Factor w/ 149 levels "ALFL","AMBI",..: 43 46 95 95 95 107 107 140 21 38 ...
 $ Count    : num  1 1 1 1 1 1 1 1 1 1 ...
 $ Behav    : Factor w/ 3 levels "C","S","V": 1 1 2 2 2 2 2 2 2 2 ...
 $ Dur      : Factor w/ 3 levels "0-3min","3-5min",..: 3 3 1 1 1 1 1 1 3 1 ...
 $ Dis      : Factor w/ 3 levels "0-50m","50-100m",..: 3 3 1 2 2 1 1 2 1 1 ...
```

```r
d |> filter(SiteID == "CL10118", SpeciesID == "OVEN")
```

```
   SiteID SpeciesID Count Behav     Dur     Dis
1 CL10118      OVEN     1     S  0-3min 50-100m
2 CL10118      OVEN     1     S  0-3min 50-100m
3 CL10118      OVEN     1     S  0-3min   100+m
4 CL10118      OVEN     1     S 5-10min   0-50m
```

We can work with a single species or multiple species. Notice that `SiteID`, `SpeciesID`, etc. are coded as factors. This allows us to keep all the `SiteID` even after subsetting. If we keep these columns as plain character, we have to remember that there are no null records, i.e. sites where the species was not detected. The long format only contains information about detections.

If for some reason, no individuals of any species were detected at a site, we would add a `"NONE"` placeholder for the `SpeciesID`.

Point count duration

Each point-count in this data set has a total duration of 10 minutes, and 3 time intervals: 0–3, 3–5, and 5–10 minutes.

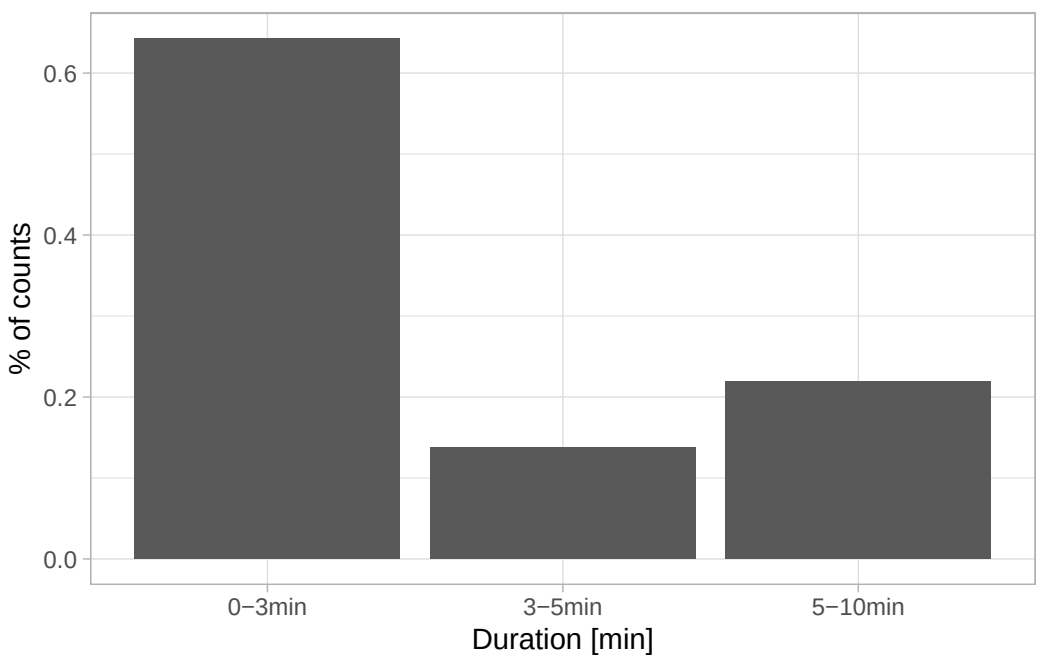
We used the `table()` function to tabulate the frequency of counts (rows) by time interval.

```r
100 * table(d$Dur) / nrow(d)
```

```
  0-3min   3-5min  5-10min
64.24807 13.78599 21.96594
```

This is similar to `tally()`:

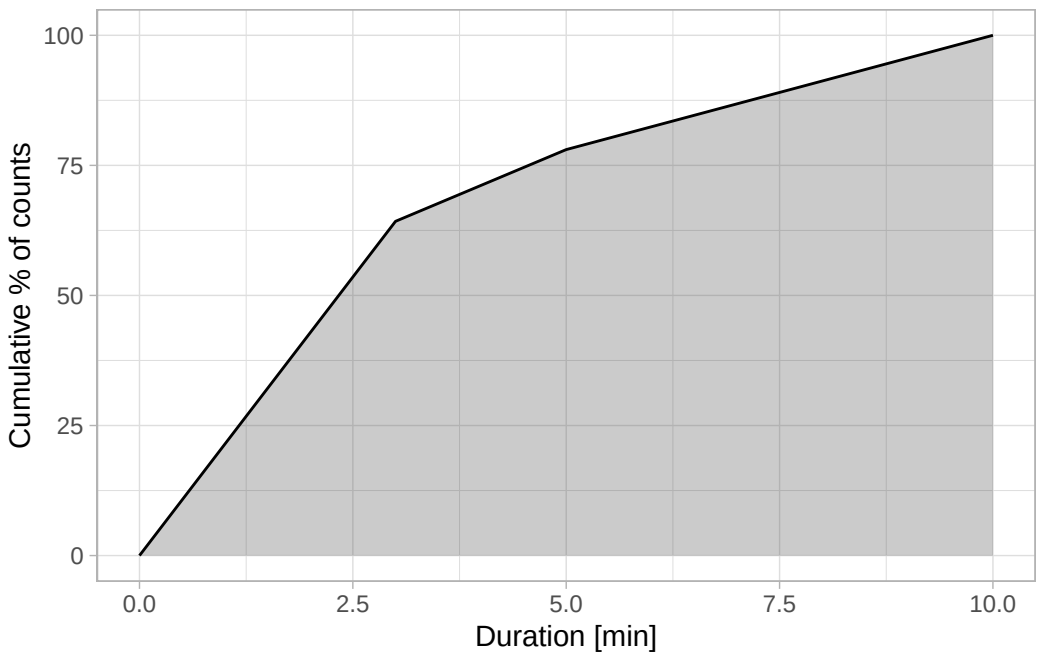
```
d |>
    group_by(Dur) |>
    tally() |>
    mutate(
        Perc = n / sum(n)
    ) |>
    ggplot(aes(x = Dur, y = Perc)) +
    geom_bar(stat = "identity") +
    xlab("Duration [min]") +
    ylab("% of counts") +
    theme_light()
```



We see most of the counts in the first 3 minutes, least of the counts in the next 2 minutes, and the rest th the second half of the total 10 minutes duration. These counts represent new individuals.

We can also plot this as a cumulative percentage:

```r
data.frame(
    Dur = c(0, 3, 5, 10),
    Cumul = c(0, 100 * cumsum(table(Dur = d$Dur) / nrow(d)))
) |>
    ggplot(aes(x = Dur, y = Cumul)) +
    # geom_bar(stat = "identity") +
    geom_area(alpha = 0.25) +
    geom_line() +
    xlab("Duration [min]") +
    ylab("Cumulative % of counts") +
    theme_light()
```



Point count radius

The data set includes 3 distance bands:

- 0–50 m
- 50–100 m
- 100 m

This is a so called unlimited distance count, that is quite common.

When the individuals are counted to an unknown distance (infinity), It is hard to define the survey area. We'll show some ways of estimating it later.

Sometimes a 400 m limit is assumed as the maximum distance from which cues can be heard, which is why BBS stations are 800 m (half mile) apart.

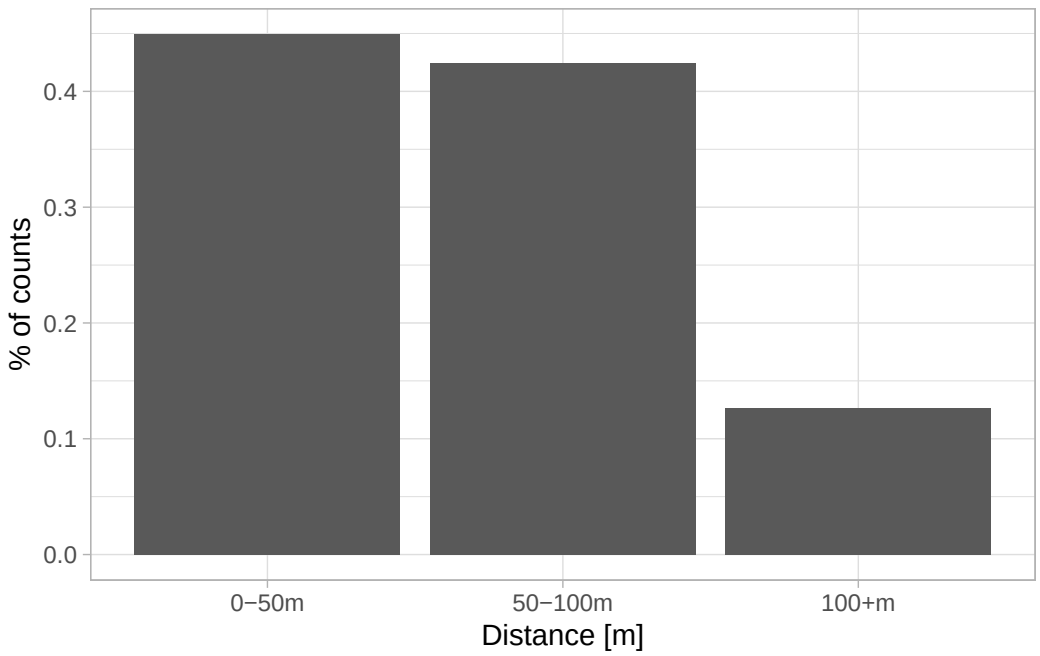
For truncated radius counts, the maximum distance within which the individuals are detected, the area is known. E.g. if we use the 0–50 m circle, the area is $A = (50m)^2 * \pi$. For the first 2 distance bands combined, $A = (100m)^2 * \pi$.

```
100 * table(d$Dis) / nrow(d)
```

```
   0-50m  50-100m    100+m 
44.93050 42.44635 12.62316 
```

Most of the counts were in the 0–50 m bin, but almost equal frequency was counted in the 50–100 m bin, which is only 3 times larger area. The smallest portion is outside of the 100 m distance.

```
d |>
    group_by(Dis) |>
    tally() |>
    mutate(Perc = n / sum(n)) |>
    ggplot(aes(x = Dis, y = Perc)) +
    geom_bar(stat = "identity") +
    xlab("Distance [m]") +
    ylab("% of counts") +
    theme_light()
```



We can also tabulate both the distance and duration bins:

```
100 * round(table(Dis = d$Dis, Dur = d$Dur) / nrow(d), 3) |>
    addmargins()
```

```
         Dur
Dis       0-3min 3-5min 5-10min   Sum
  0-50m     30.7    5.3     8.9  44.9
  50-100m   27.0    6.1     9.4  42.5
  100+m      6.5    2.4     3.7  12.6
  Sum       64.2   13.8    22.0 100.0
```

Working with a single species

Working with subsets of the data frame using Tidyverse functions can be misleading because non-detections are not explicitly stored.

```
y1 <- d |>
    filter(SpeciesID == "OVEN") |>
    group_by(SiteID) |>
    summarize(y = n())
dim(y1)
```

```
[1] 2077    2
```

Better to use functions that keep site ID's for non-detections. For example the `Xtab()` function from the mefa4 package.

```
y2 <- mefa4::Xtab(Count ~ SiteID + SpeciesID, d)[, "OVEN", drop = FALSE]
dim(y2)
```

```
[1] 4569    1
```

Let's see the total counts by distance and duration intervals:

```
y3 <- mefa4::Xtab(Count ~ Dis + Dur, d, subset = d$SpeciesID == "OVEN")
100 * y3 / sum(y3)
```

```
3 x 3 sparse Matrix of class "dgCMatrix"
          0-3min   3-5min  5-10min
0-50m   27.47036 2.692688 3.162055
50-100m 42.09486 6.052372 7.411067
100+m    6.59585 1.926877 2.593874
```

It is also possible to tabulate counts in a 3-way fashion, e.g. calculate the distance x duration combinations for each site. The first 2 dimensions will be treated as rows and columns of matrices. The 3rd dimension is used is used to form a named list.

```
y4 <- mefa4::Xtab(Count ~ Dis + Dur + SiteID, d, subset = d$SpeciesID == "OVEN")
y4[1:2]
```

```
$CL10102
3 x 3 sparse Matrix of class "dgCMatrix"
        0-3min 3-5min 5-10min
0-50m        1      .       .
50-100m      2      .       .
100+m        .      .       .

$CL10106
3 x 3 sparse Matrix of class "dgCMatrix"
        0-3min 3-5min 5-10min
0-50m        .      .       .
50-100m      .      .       .
100+m        .      .       .
```

Slightly more useful (and faster) to use site IDs as the 1st dimension, though:

```
y5 <- mefa4::Xtab(Count ~ SiteID + Dis + Dur, d, subset = d$SpeciesID == "OVEN")
lapply(y5, head, 2)
```

```
$`0-3min`
2 x 3 sparse Matrix of class "dgCMatrix"
        0-50m 50-100m 100+m
CL10102     1       2     .
CL10106     .       .     .

$`3-5min`
2 x 3 sparse Matrix of class "dgCMatrix"
        0-50m 50-100m 100+m
CL10102     .       .     .
CL10106     .       .     .

$`5-10min`
2 x 3 sparse Matrix of class "dgCMatrix"
        0-50m 50-100m 100+m
CL10102     .       .     .
CL10106     .       .     .
```

We can calculate different sums:

```
# 3min 50m
table(y5[["0-3min"]][, "0-50m"])
```

```
   0    1    2    3    4
3654  737  161   15    2
```

```
# 5min 100m
table(rowSums(y5[["0-3min"]][, 1:2]) + rowSums(y5[["3-5min"]][, 1:2]))
```

```
   0    1    2    3    4    5    6
2784  846  603  246   73   14    3
```

```r
# 10min Inf
table(rowSums(y5[[1]]) + rowSums(y5[[2]]) + rowSums(y5[[3]]))
```

```
   0    1    2    3    4    5    6
2492  881  654  365  134   30   13
```

Working with multiple species

To get usable species data, we need to pivot the data. The crosstabulation uses sparse matrices (0's are not stored), which leads to smaller memory footprint and increased speed. This approach exploits the fact that fill rate (proportion of non-zero values) of bird count data is pretty low (~5% in this case).

```r
ytot <- mefa4::Xtab(Count ~ SiteID + SpeciesID, d)

head(ytot[, 1:10])
```

```
6 x 10 sparse Matrix of class "dgCMatrix"

  [[ suppressing 10 column names 'ALFL', 'AMBI', 'AMCO' ... ]]


CL10102 . . . . . . . . . .
CL10106 . . . . . . . . . .
CL10108 2 . . . . . . . . .
CL10109 . . . 2 . . . . . .
CL10111 . . . . . . 1 . . .
CL10112 . . . . . . . . . .
```

```r
mean(ytot > 0)
```

```
[1] 0.05014388
```

A sparse matrix can be converted to ordinary matrix

```r
head(as.matrix(ytot)[, 1:10])
```

```
        ALFL AMBI AMCO AMCR AMGO AMKE AMRE AMRO AMWI ATTW
CL10102    0    0    0    0    0    0    0    0    0    0
CL10106    0    0    0    0    0    0    0    0    0    0
CL10108    2    0    0    0    0    0    0    0    0    0
CL10109    0    0    0    2    0    0    0    0    0    0
CL10111    0    0    0    0    0    0    1    0    0    0
CL10112    0    0    0    0    0    0    0    0    0    0
```

Check if counts are as expected:

```r
max(ytot) # this is interesting
```

```
[1] 200
```

```r
rev(sort(apply(as.matrix(ytot), 2, max)))[1:10] # it is CAGO
```

```
CAGO BLTE RECR PISI WWCR CEDW BLBW DCCO GRAJ AMPI
 200   70   51   50   40   23   20   19   16   12
```

```r
## flyover (FO) flock (FL) beyond 100m distance
detect::josm$counts |>
    filter(
        SiteID == rownames(ytot)[which(ytot[, "CAGO"] == 200)],
        SpeciesID == "CAGO"
    ) |>
    head(2)
```

```
                        ObservationID  SiteID StationID TimeInterval Direction
CO10712-130603-008 CO10712-130603-008 CO10712 CO10712-1            1         2
CO10712-130603-009 CO10712-130603-009 CO10712 CO10712-1            1         2
                    Distance DetectType1 DetectType2 DetectType3 Sex Age
CO10712-130603-008        3           C        <NA>        <NA>   U   A
CO10712-130603-009        3           C        <NA>        <NA>   U   A
                    Activity1 Activity2 Activity3 ActivityNote    Dur   Dis
CO10712-130603-008        FO        FL      <NA>         <NA> 0-3min 100+m
CO10712-130603-009        FO        FL      <NA>         <NA> 0-3min 100+m
                    SpeciesID
CO10712-130603-008      CAGO
CO10712-130603-009      CAGO
```

The nice thing about this cross tabulation is that we can filter the records without changing the structure (rows, columns) of the table:

```
ytot_seen <- Xtab(Count ~ SiteID + SpeciesID, d, subset = d$Behav == "V")
ytot_heard <- Xtab(Count ~ SiteID + SpeciesID, d, subset = d$Behav != "V")
```

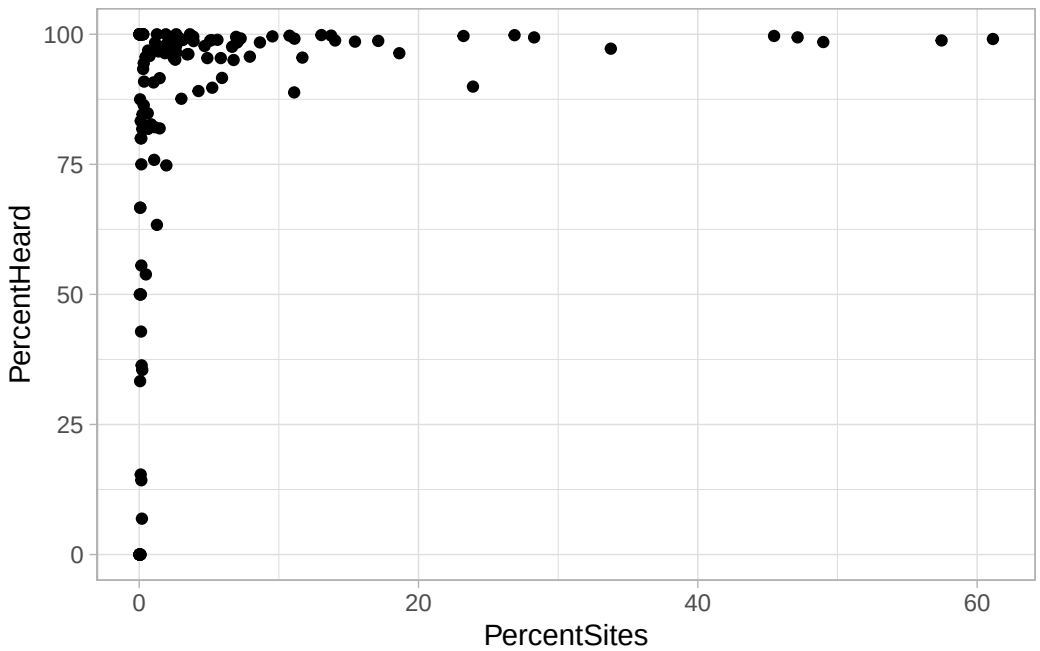
We see that the species that are detected only via visual cues are rare, mostly raptors and waterfowls:

```
ph <- data.frame(
    PercentSites = 100 * colSums(ytot > 0) / nrow(ytot),
    PercentHeard = 100 * colSums(ytot_heard) / colSums(ytot),
    SpeciesID = colnames(ytot)
) |>
    arrange(PercentHeard) |>
    left_join(detect::josm$species, by = join_by(SpeciesID))

head(ph)
```

```
  PercentSites PercentHeard SpeciesID              SpeciesName
1   0.08754651            0      BUFF               Bufflehead
2   0.08754651            0      BWTE         Blue-winged Teal
3   0.04377325            0      COGO         Common Goldeneye
4   0.02188663            0      COHA            Cooper's Hawk
5   0.04377325            0      DCCO Double-crested Cormorant
6   0.08754651            0      GWTE        Green-winged Teal
         ScientificName
1     Bucephala albeola
2          Anas discors
3    Bucephala clangula
4    Accipiter cooperii
5 Phalacrocorax auritus
6           Anas crecca
```

```
ph |>
    ggplot(aes(x = PercentSites, y = PercentHeard)) +
    geom_point() +
    theme_light()
```



The columns in these matrices contain the species counts, and will serve as the response variable (also called dependent variable) in the models that we fit next.